

Restorations in Radiation: Material Choices for Challenging Cases

Group 2

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Epidemiology of Oral Cancer

Prevalence:

- Oral cancer affects nearly **390,000 people worldwide each year**,
- In the United States: **45,703 new cases** and **10,835 deaths** annually in 2020.
- Highest incidence: in **males, smokers, and alcohol consumers**.

Most Common Type:

- The majority (~90%) of oral cancers are oral squamous cell carcinoma (OSCC), of the mouth and oropharynx.



Epidemiology of Oral Cancer

Treatments:

- Surgery
- Radiotherapy (RT)
- Chemotherapy (often combined with RT in advanced cases)



Dental Relevance

Early Detection:

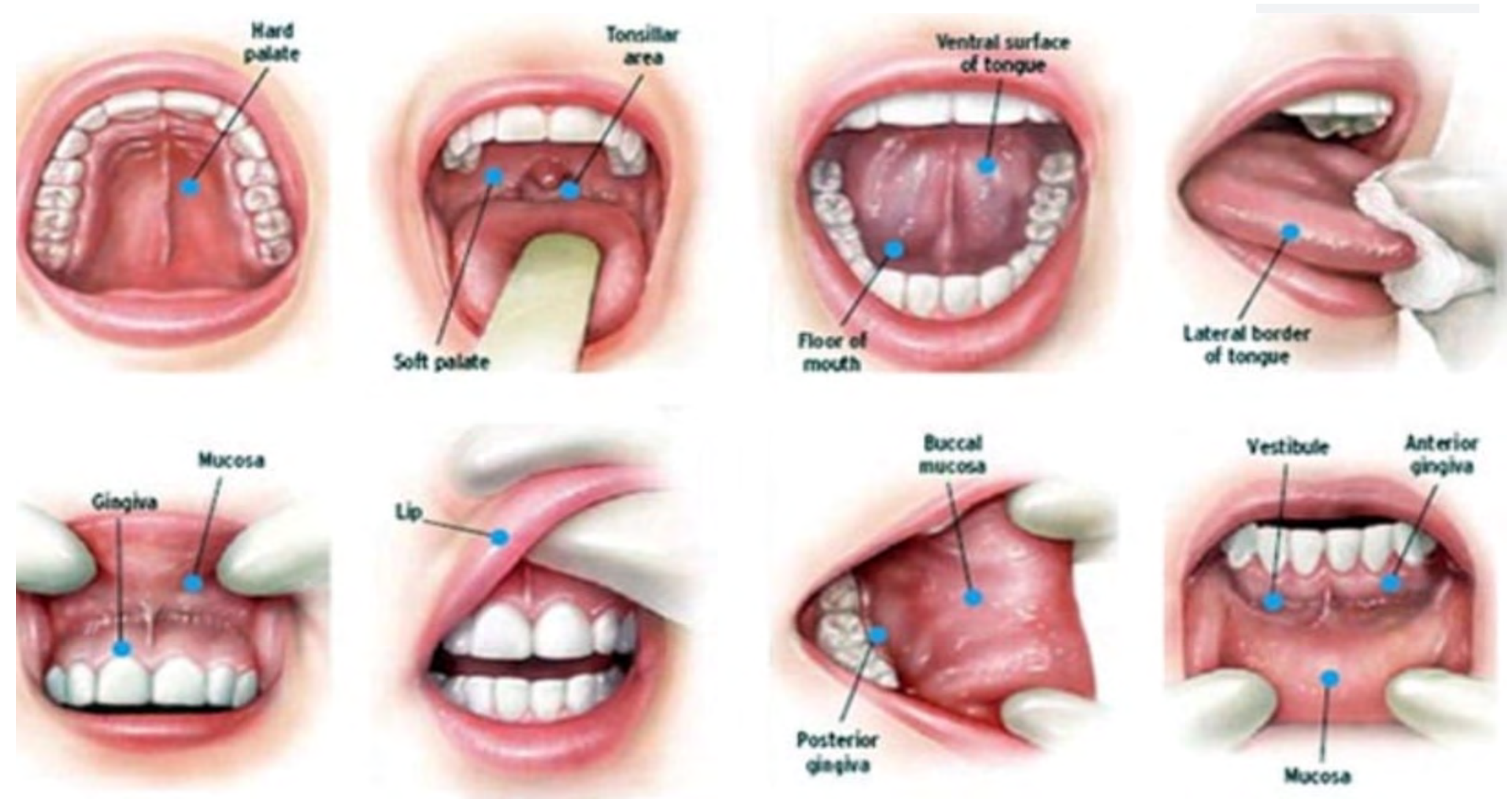
- Dentists can identify lesions or premalignant changes

Radiation Effects:

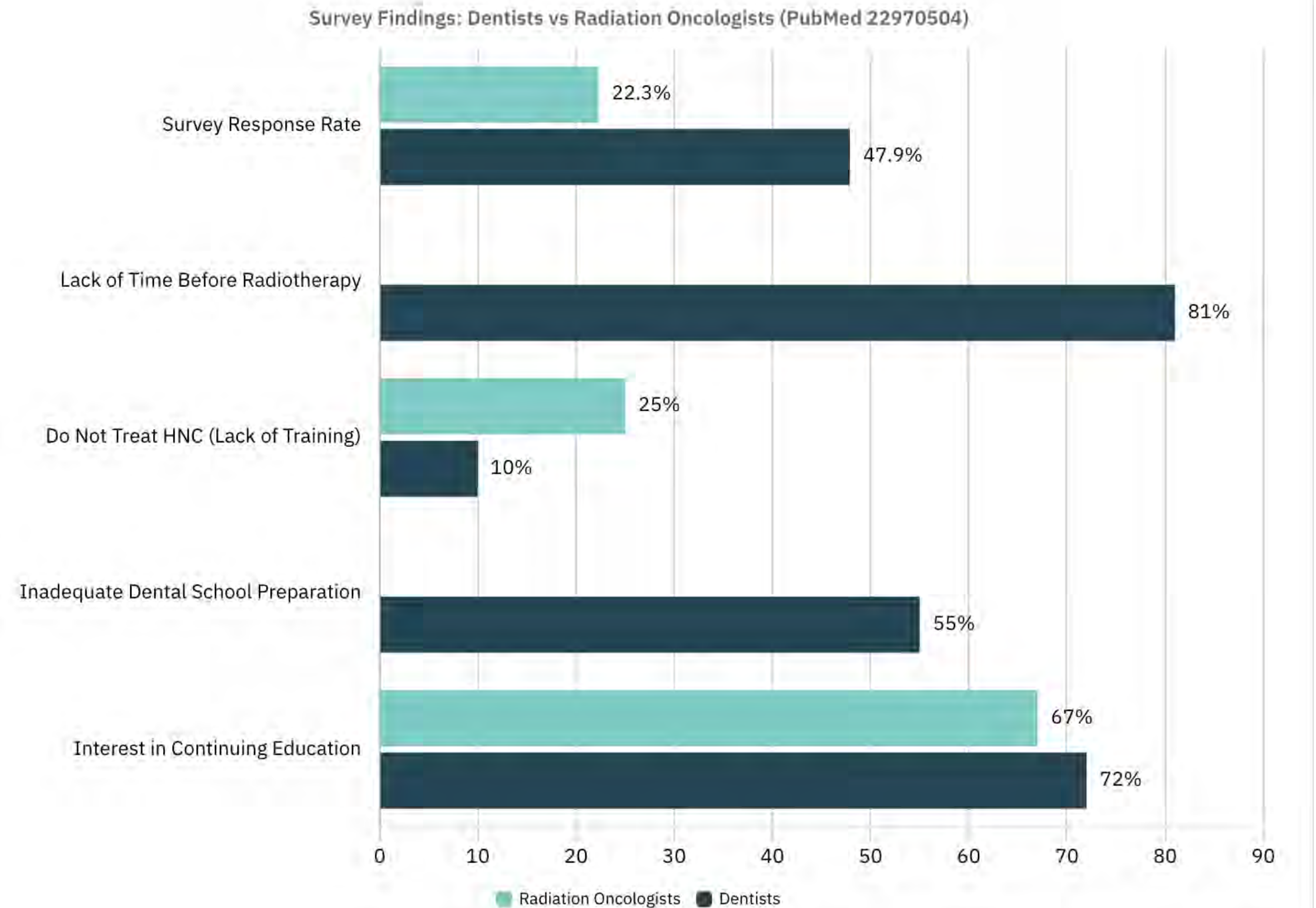
- Compromised teeth and bone

Restorative Challenges:

- Choose restorative materials that maximize durability and longevity



Survey of Michigan dentists and radiation oncologists on oral care of patients undergoing head and neck radiation therapy.



Identify the oral and systemic effects of head and neck radiation therapy, including its impact on teeth, bonding substrates, salivary function, and overall restorative prognosis.

Doha Akid and Mildred Bonful

Search Strategy



P: Patients who have received **head and neck radiation therapy** and are experiencing changes in oral tissues.

I: Evaluation of the effects of radiation therapy on oral structures including teeth, bonding substrates, and salivary glands and their impact on restorative dental outcomes.

C: Patients **without head and neck radiation exposure** or with **non-irradiated teeth and tissues**.

O: Identification of **oral and systemic effects**

Keywords

restorative materials, head and neck radiation

Pubmed search string

((Restorative materials) AND (head and neck radiation))

Search results

154 results
36 selected
8 used

Databases used

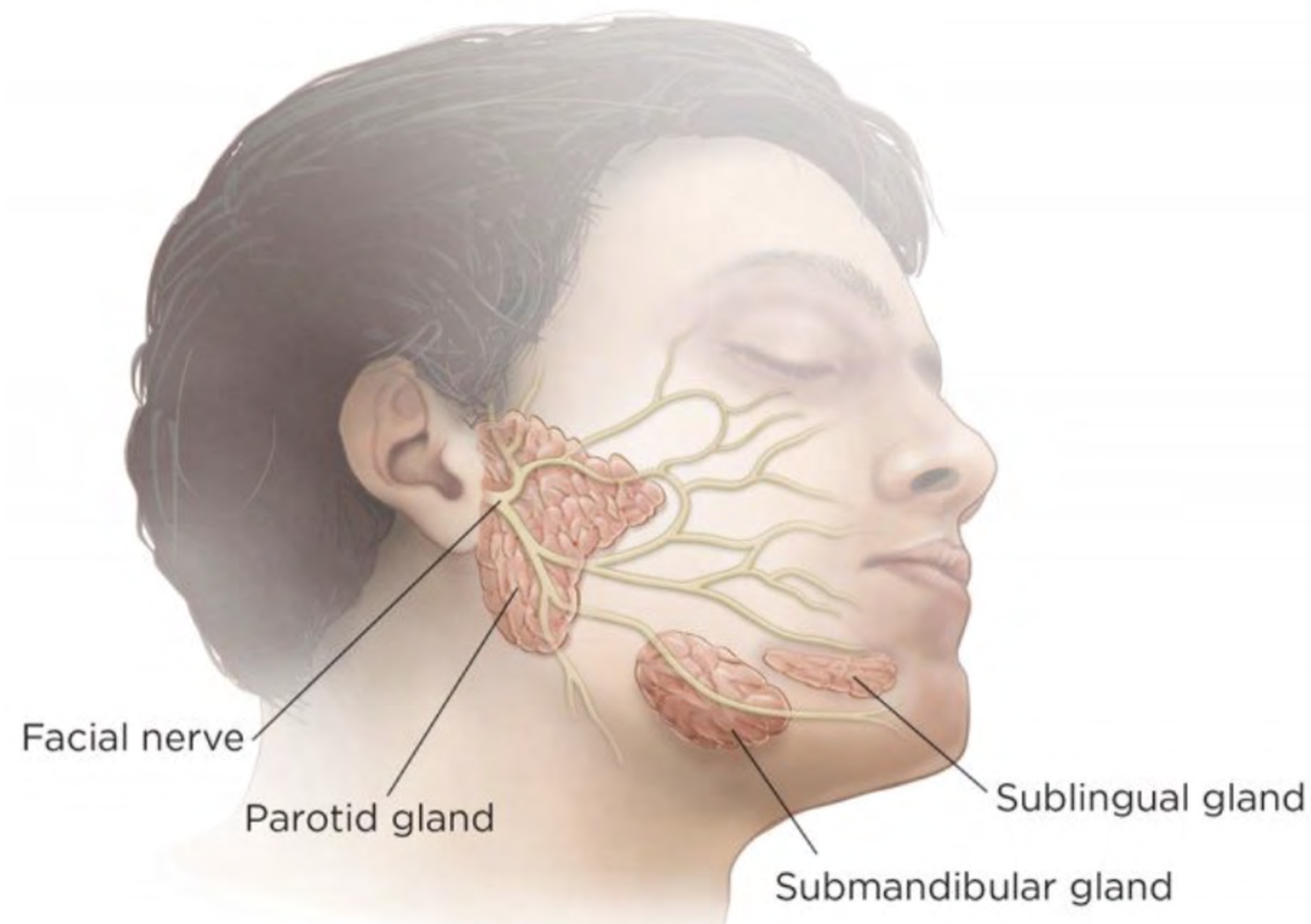
Pubmed

Highest level of evidence

Systematic reviews, systematic reviews of randomized clinical trials

Oral and Systemic Effects

- Damages to salivary glands leading to xerostomia and radiation-related caries



Conventional vs. Intensity-Modulated RT

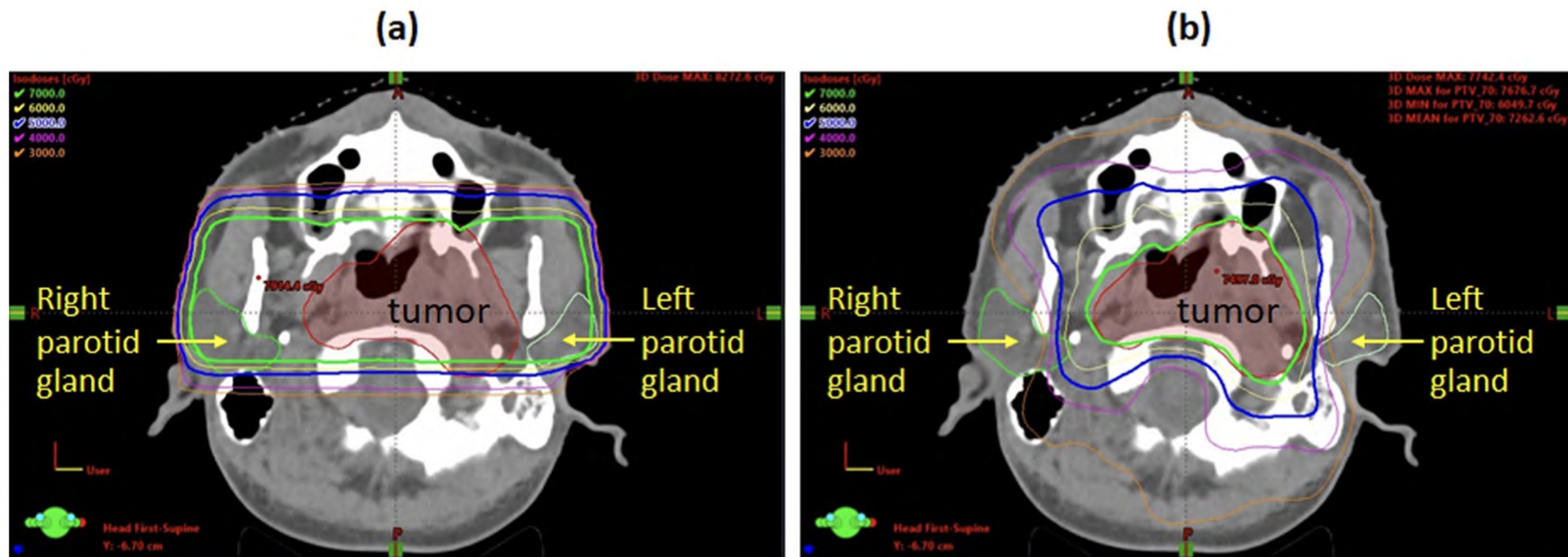
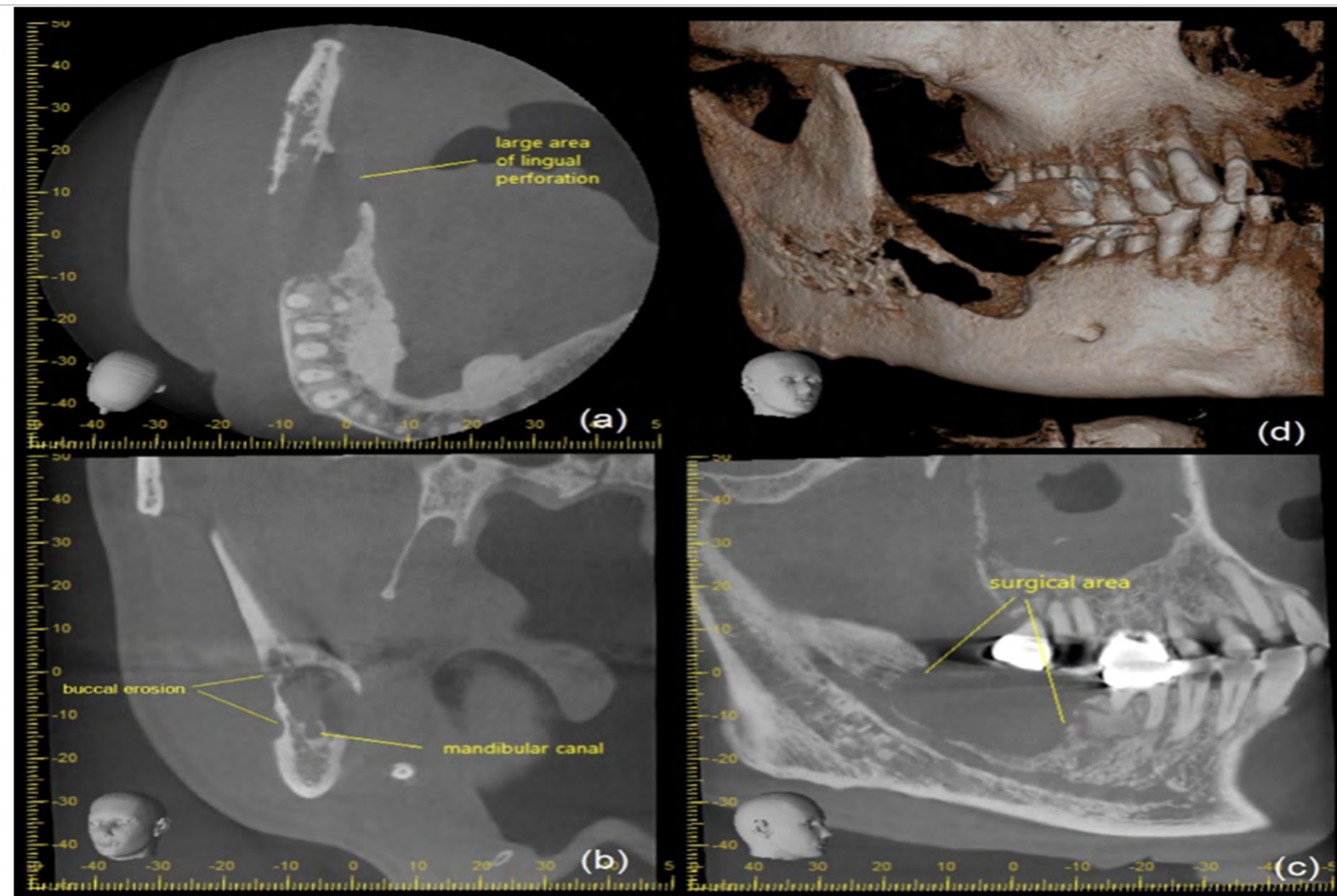
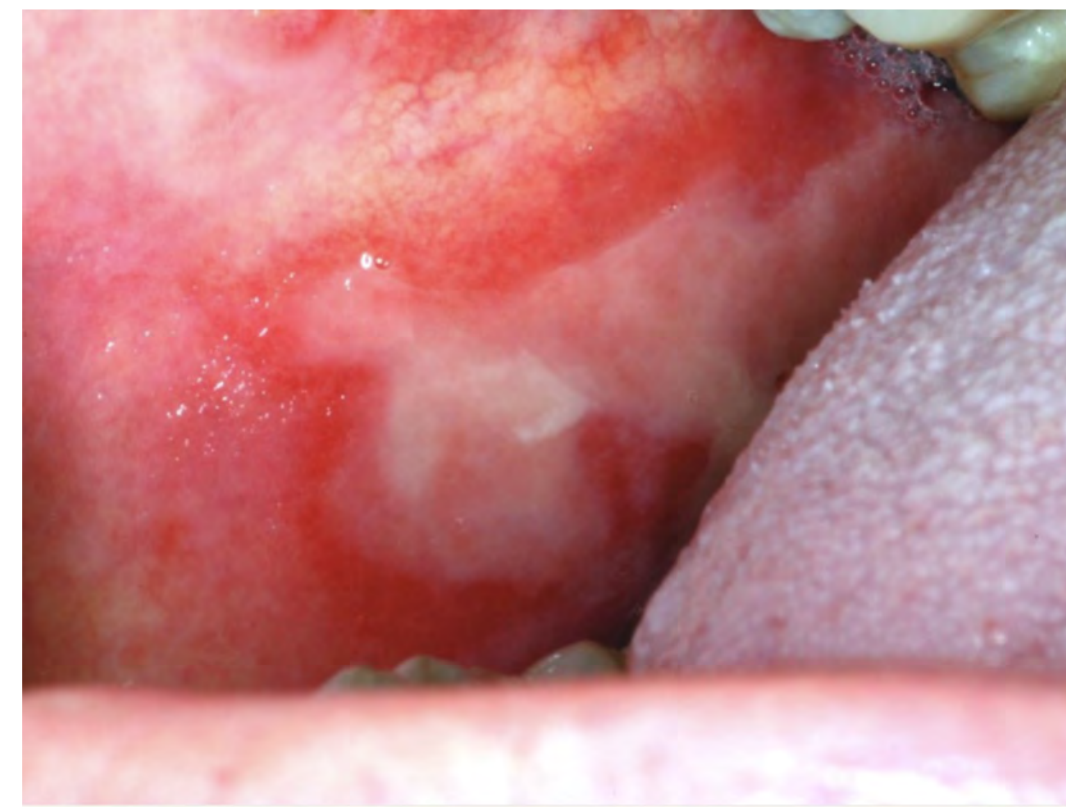


Figure 2 Radiation dose distribution with the conventional radiation technique (a) and intensity-modulated radiotherapy (IMRT) technique (b). Radiation doses of 50 Gy (green line) and 70 Gy (blue line) to the microscopic (subclinical) and gross tumor volumes, respectively. The recommended median dose to the parotid glands is <30 Gy (orange line), and these glands are better spared by the use of the IMRT technique

Oral and Systemic Effects

- Oral Mucositis
- Oral Candidiasis
- Lymphedema
- Trismus
- Osteoradionecrosis
- Periodontitis



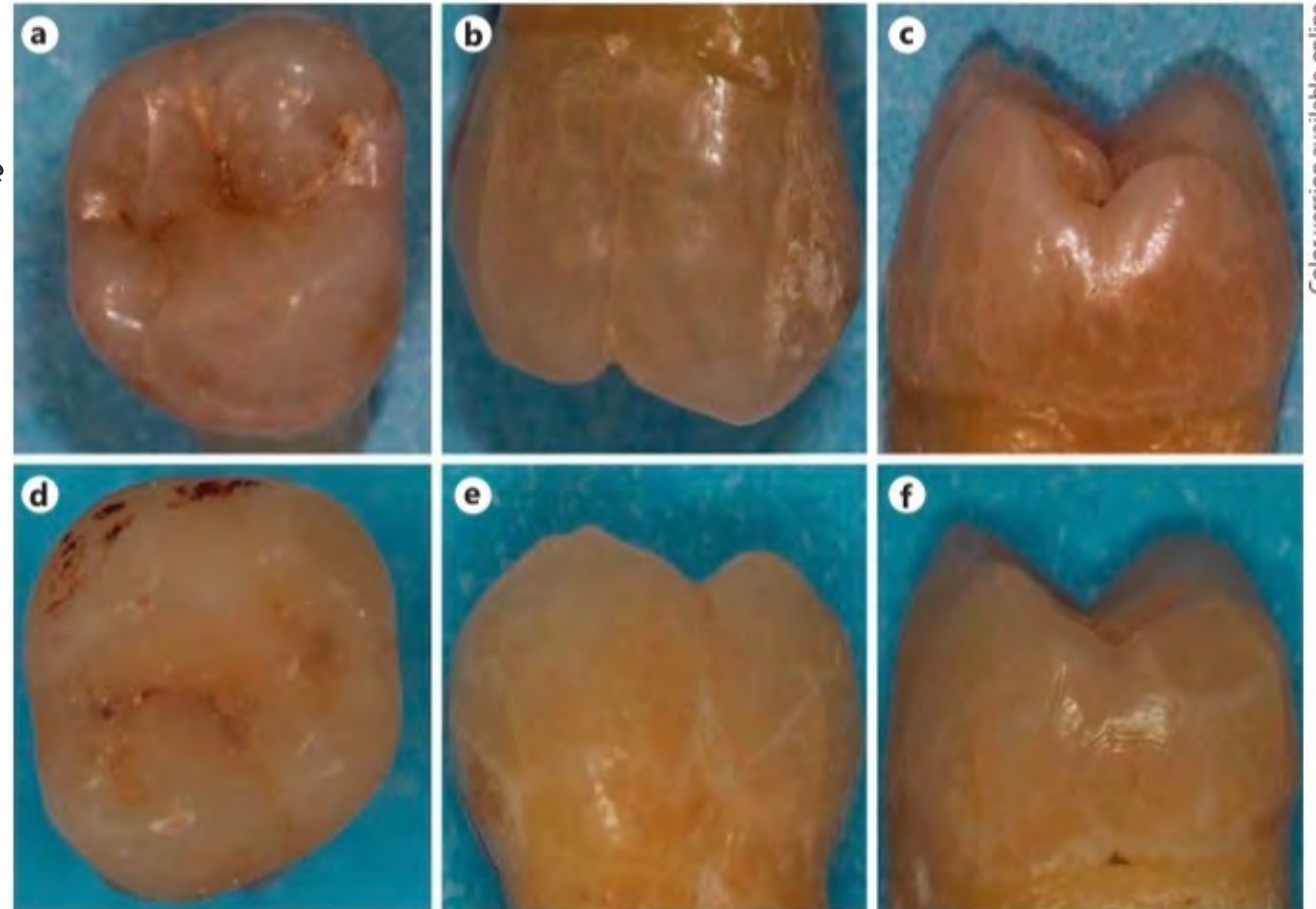
Structural Analysis of Enamel in Teeth from Head-and-Neck Cancer Patients Who Underwent Radiotherapy

Cristhian C. Madrid^a Mariana de Pauli Paglioni^b Sergio R. Line^c
Karina G. Vasconcelos^d Thaís Bianca Brandão^d Marcio A. Lopes^b
Alan Roger Santos-Silva^{b,d} Mario Fernando De Goes^a

- No qualitative differences observed in both groups.
 - Enamel showed some microcracks, wear, and normal pigmentation.
 - Buccal and lingual surfaces had normal textures and pigmentation
 - No clinical signs of caries in experimental samples

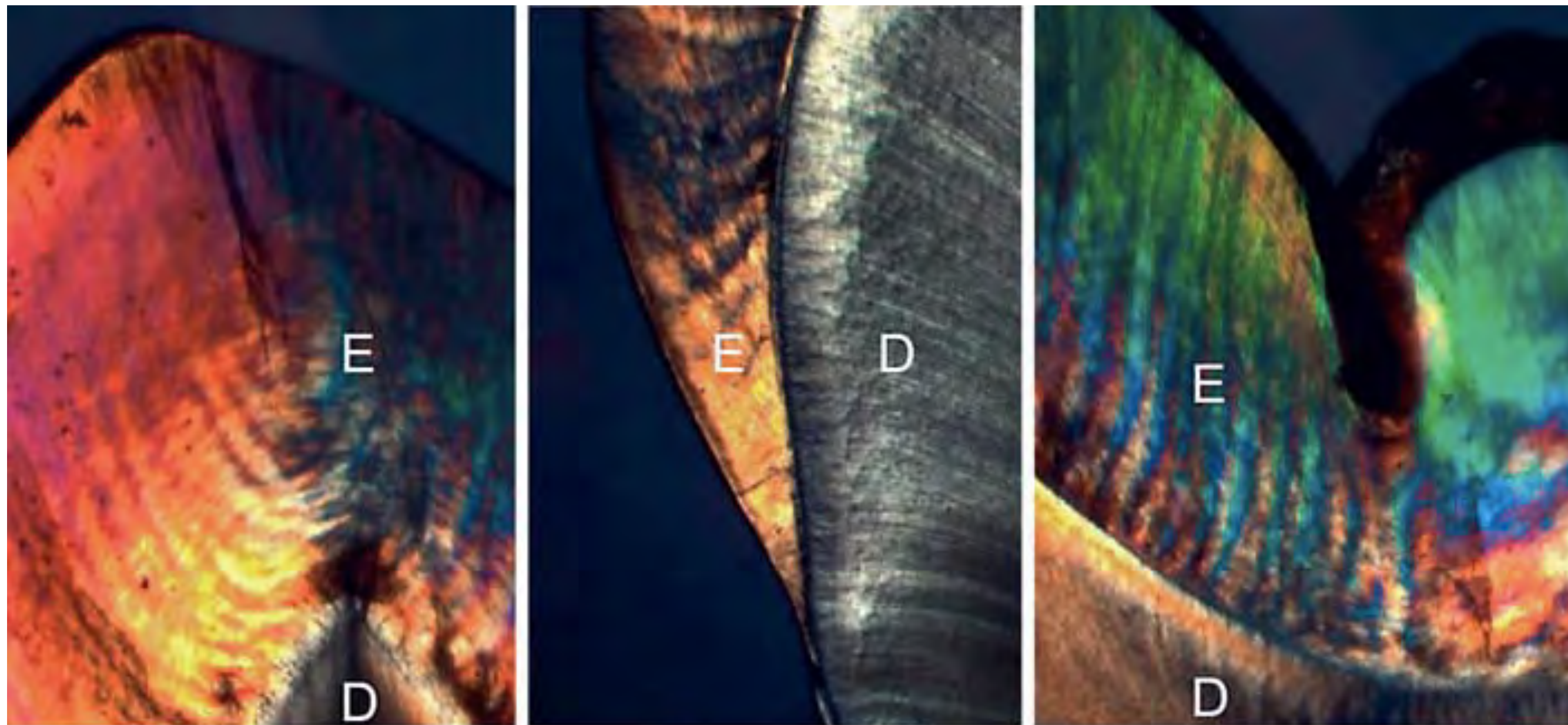
Control group

Irradiated group

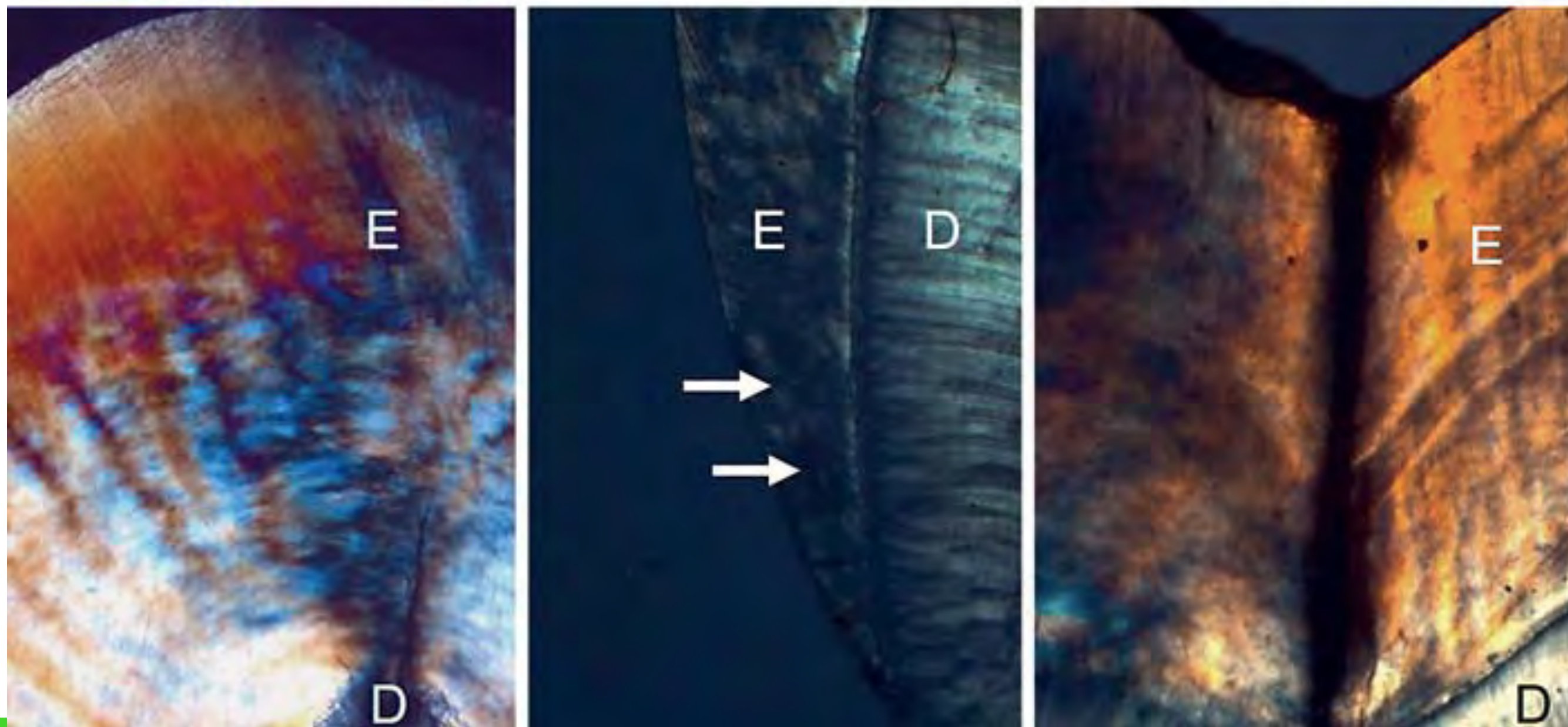


Polarized Light Microscopy

Control group



Irradiated group



- Both groups showed strong, colorful birefringence (yellow, blue, red, orange) in enamel cusps.
 - Striae of Retzius and enamel tufts were visible in both groups.
- Irradiated cervical enamel showed reduced birefringence (black to gray shades).
 - Control cervical enamel remained bright and similar to occlusal enamel.
- Hunter-Schreger bands were clearly seen in both groups.
- Occlusal pit enamel had similar color and brightness in both groups.



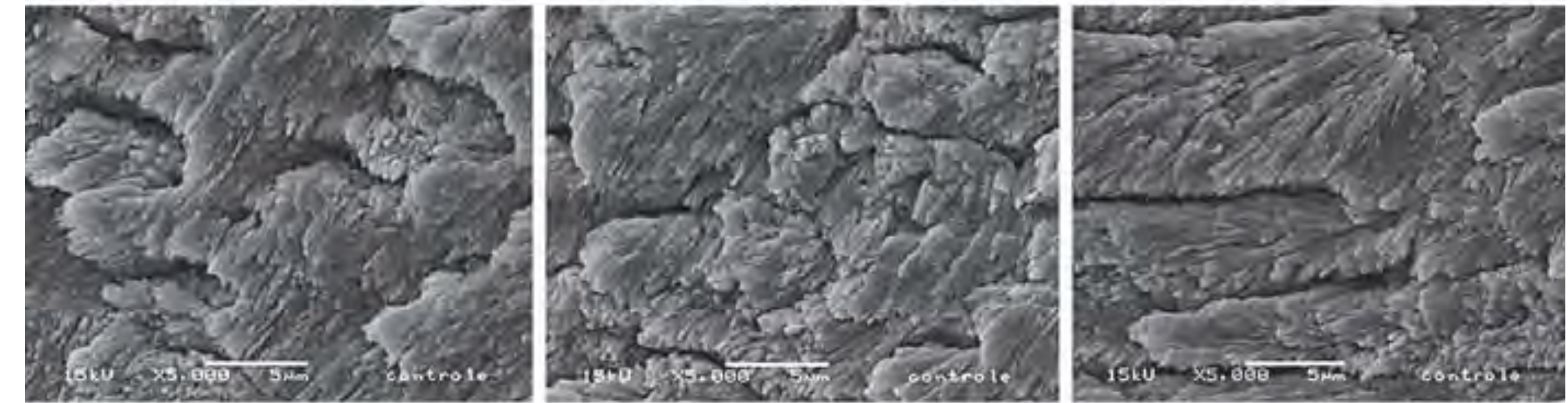
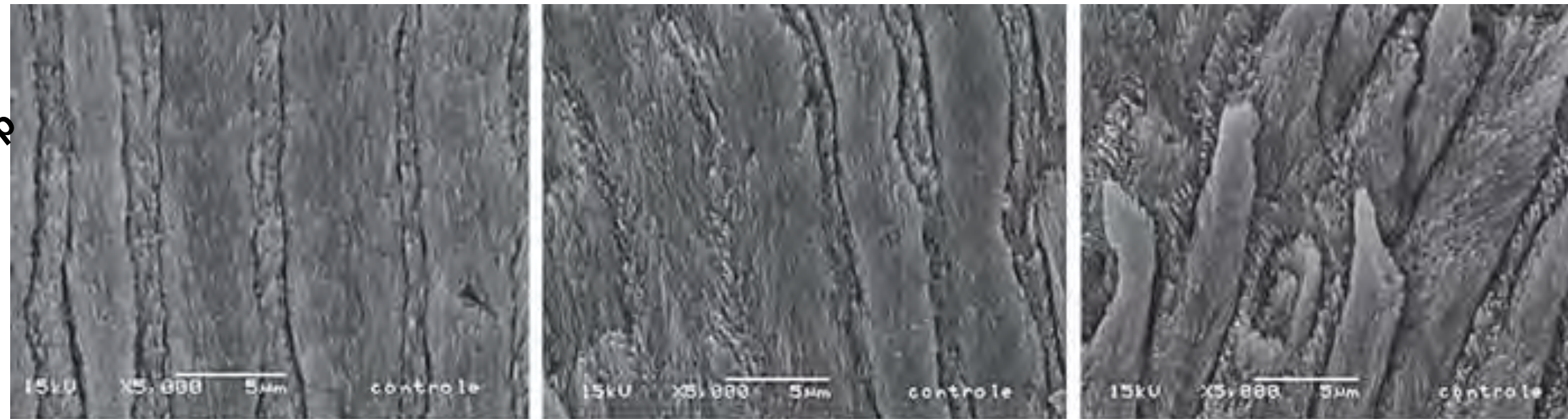
Scanning Electron Microscopy



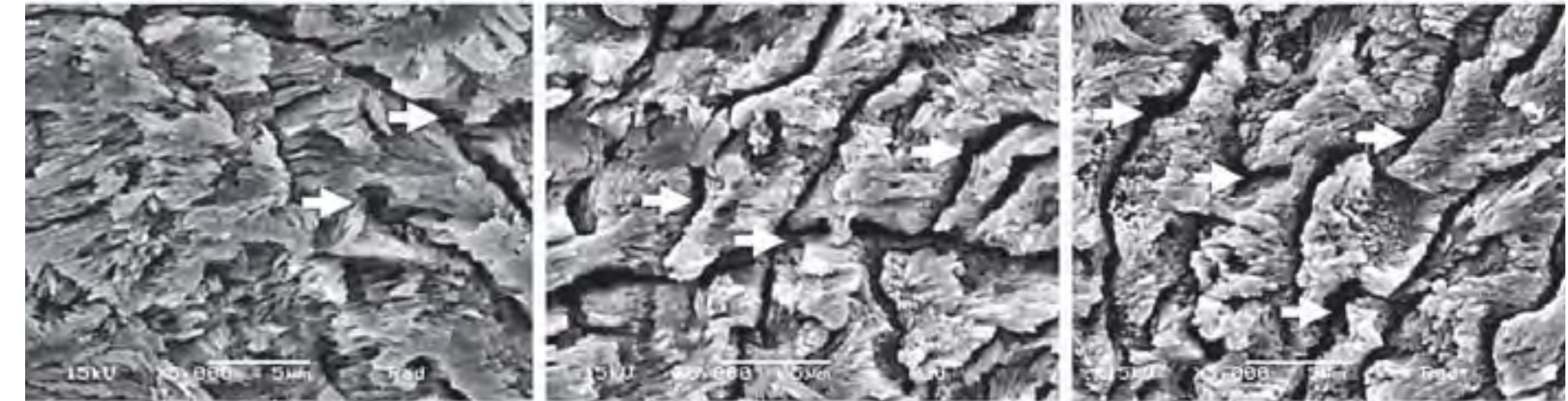
Enamel from the cusp region

Enamel from the cervical region

Control group



Irradiated group



Irradiation caused widened interprismatic spaces, especially in the inner enamel of the cusp and cervical regions which are more prone to radiation caries.

Table 1. Remnant depth and width of scratches on the enamel before and after irradiation under different loads in relation to prism orientation.

	The perpendicular-sectioned enamel slides			The parallel-sectioned enamel slides		
	Before irradiation (n = 6)	After irradiation (n = 6)	P value	Before irradiation (n = 6)	After irradiation (n = 6)	P value
Depth (nm) 20 mN	25 ± 3	29 ± 6	0.34	43 ± 18	52 ± 16	0.45
40 Mn	79 ± 8	101 ± 13	0.03	91 ± 22	154 ± 19	0.005
60 mN	181 ± 10	223 ± 10	0.0009	247 ± 74	381 ± 30	0.02
Width (um) 20 mN	2.8 ± 0.3	2.9 ± 0.2	0.55	3.7 ± 0.4	4.0 ± 0.7	0.36
40 mN	3.5 ± 0.3	3.8 ± 0.2	0.16	4.1 ± 0.5	4.4 ± 0.3	0.27
60 mN	4.7 ± 0.5	5.0 ± 0.3	0.12	4.9 ± 1.4	5.3 ± 0.7	0.56

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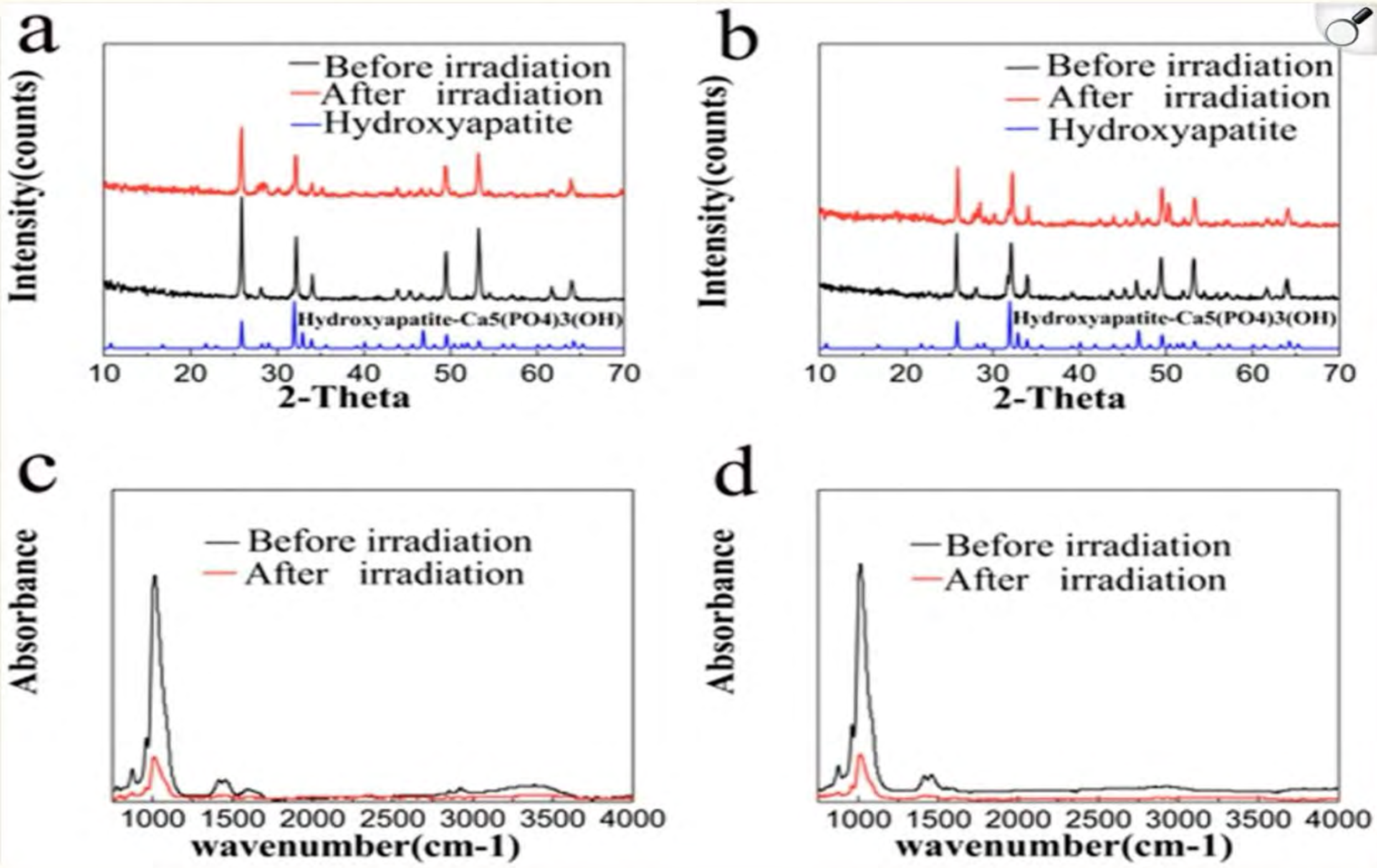
Effect of gamma irradiation on the wear behaviour of human tooth enamel

[Ping Qing](#)^{1,*}, [Shengbin Huang](#)^{2,*}, [ShanShan Gao](#)¹, [LinMao Qian](#)³, [HaiYang Yu](#)^{1,a}

❑ Radiation causes more cracks, debris, and deeper scratches.

XRD & FTIR Spectra Analysis

Figure 3. XRD patterns of enamel before and after irradiation.



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Perpendicular-sectioned enamel (a); Parallel-sectioned enamel (b). FT-IR spectrum of enamel before and after irradiation. Perpendicular-sectioned enamel (c); Parallel-sectioned enamel (d).

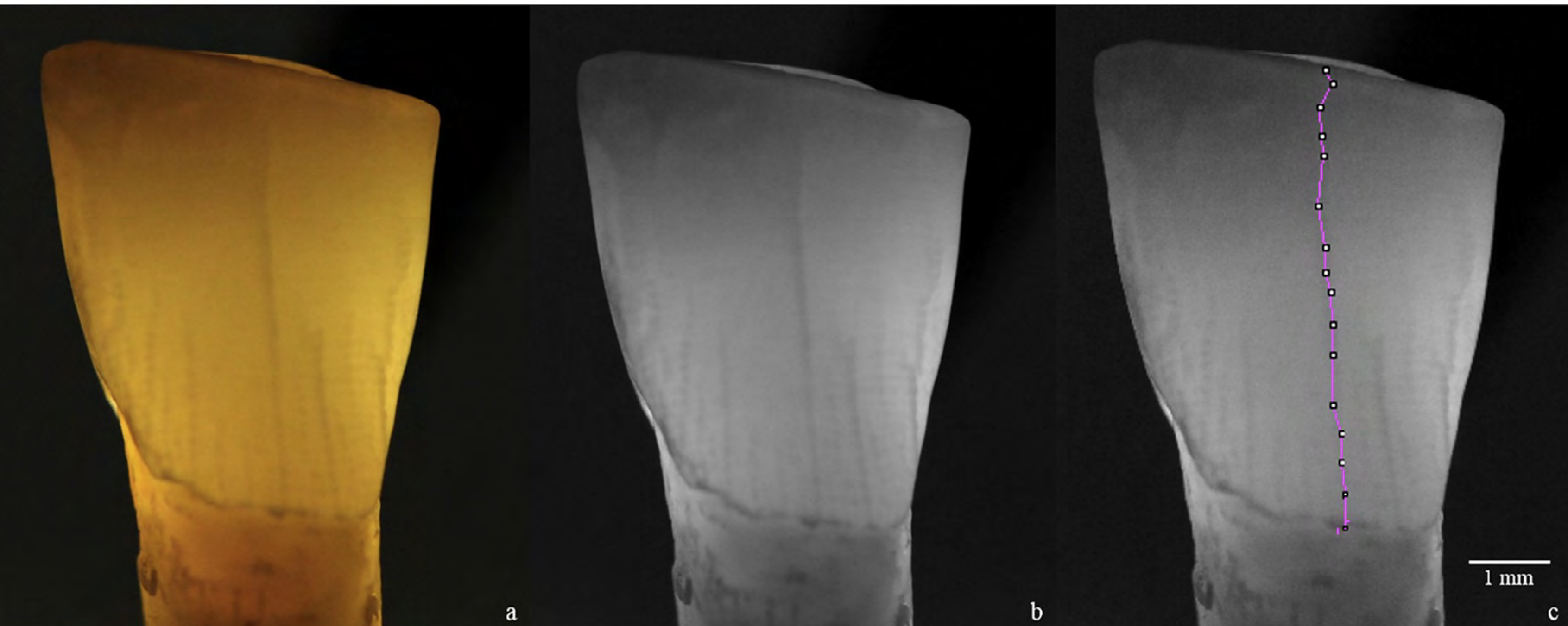
Table 2. The crystallinity, crystal size and C: M value of enamel before and after irradiation in relation to prism orientation.

	The perpendicular-sectioned enamel slides			The parallel-sectioned enamel slides		
	Before irradiation (n = 6)	After irradiation (n = 6)	P value	Before irradiation (n = 6)	After irradiation (n = 6)	P value
Crystallinity (%)	82 ± 5	74 ± 5	0.02	79 ± 8	67 ± 7	0.01
Crystal size (nm)	24 ± 1	27 ± 2	0.02	24 ± 1	27 ± 2	0.02
C:M ratio	0.034 ± 0.007	0.051 ± 0.012	0.01	0.028 ± 0.005	0.033 ± 0.003	0.02

- ❑ XRD confirmed that both groups' enamel was composed of hydroxyapatite. Irradiation decreased crystallinity and increased crystal size.
- ❑ FTIR showed phosphate and carbonate absorption decreased, with phosphate decreasing more after radiation. The carbonate-to-mineral ratio increased after irradiation.

Cracked tooth syndrome in irradiated patients with head and neck cancer

Natalia Rangel Palmier ¹, Cristhian Camilo Madrid ², Mariana de Pauli Paglioni ¹, César Rivera ³,
Beatriz Nascimento F Lebre Martins ¹, Anna Luíza Damaceno Araújo ¹, João Victor Salvajoli ⁴,
Mario Fernando de Goes ⁵, Marcio Ajudarte Lopes ¹, Ana Carolina Prado Ribeiro ⁶,
Thaís Bianca Brandão ⁷, Alan Roger Santos-Silva ⁸



- ❑ **More Enamel Craze Lines** were seen in irradiated teeth, mainly at the **cervical area**.
- ❑ Likely due to **enamel delamination** and **dentin exposure** to **highly cariogenic environment**.
- ❑ Could promote the onset and progression of **radiation-related caries (RRC)**.

The study aimed to examine the presence and patterns of enamel craze lines in teeth exposed to radiation.

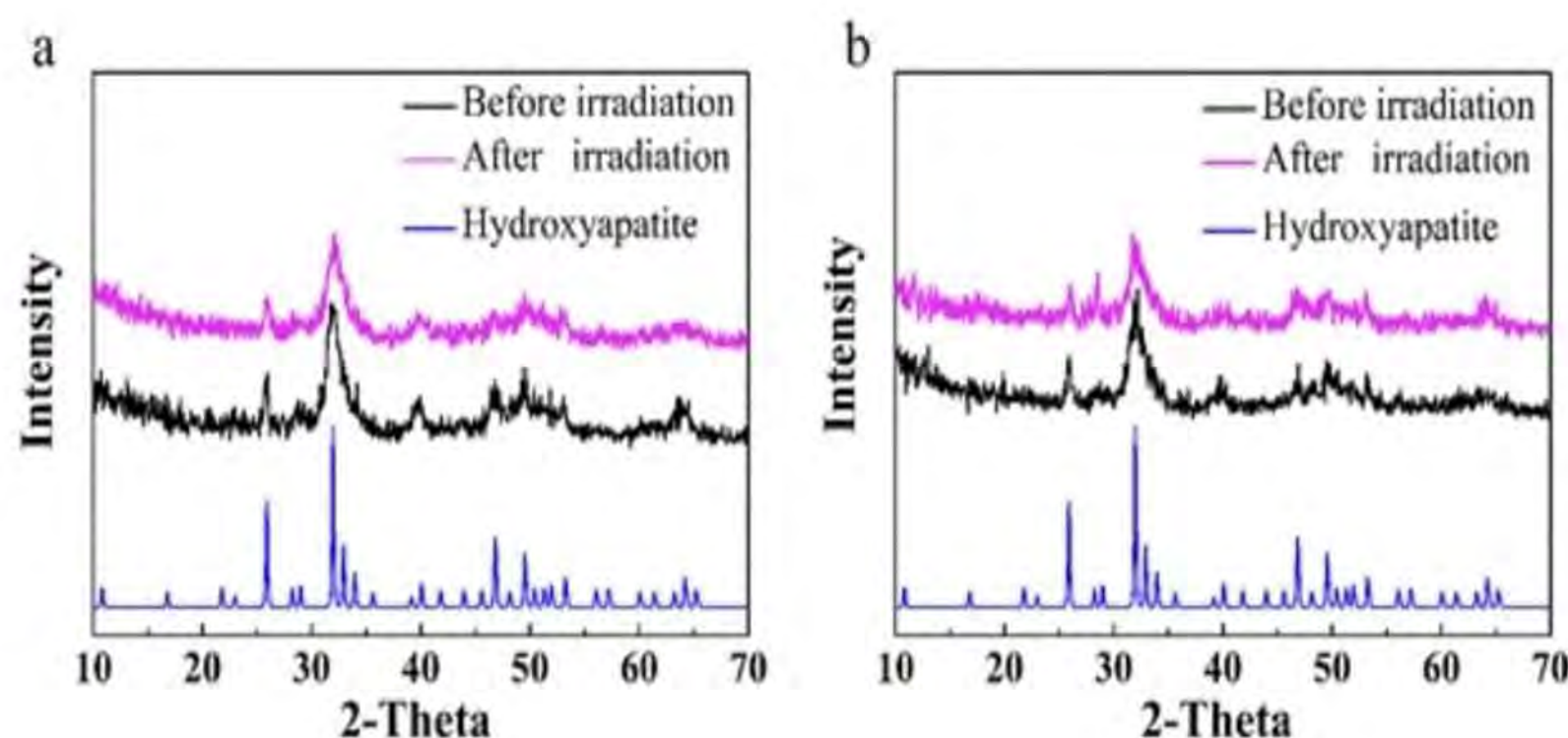
Effect of gamma irradiation on the wear behavior of human tooth dentin

Ping Qing ¹, Shengbin Huang ², ShanShan Gao ¹, LinMao Qian ³, HaiYang Yu ⁴

XRD results

Figure 5a, b presents the XRD spectra of dentin specimens cut perpendicular or parallel to tubule orientation before and after irradiation, respectively. The XRD analysis indicated that the crystalline phase in dentin before and after irradiation was also hydroxyapatite just as some previous studies described [21]. Meanwhile, the reduction in crystallinity and the increase in crystal size caused by irradiation can be seen in both the dentin cut perpendicular or parallel to tubule orientation (Table 2).

Fig. 5

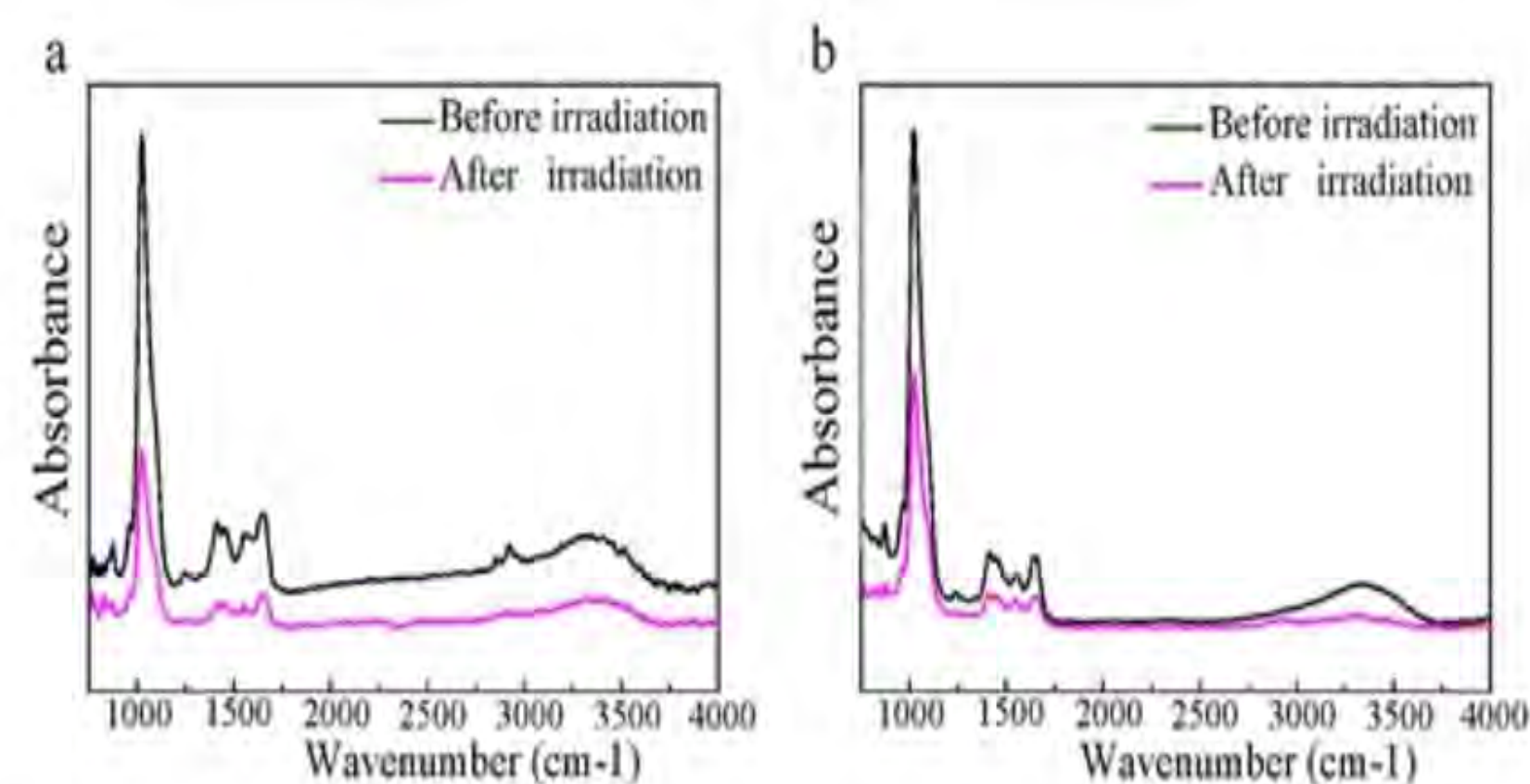


XRD spectra of dentin specimens cut perpendicular or parallel to tubule orientation before and after irradiation

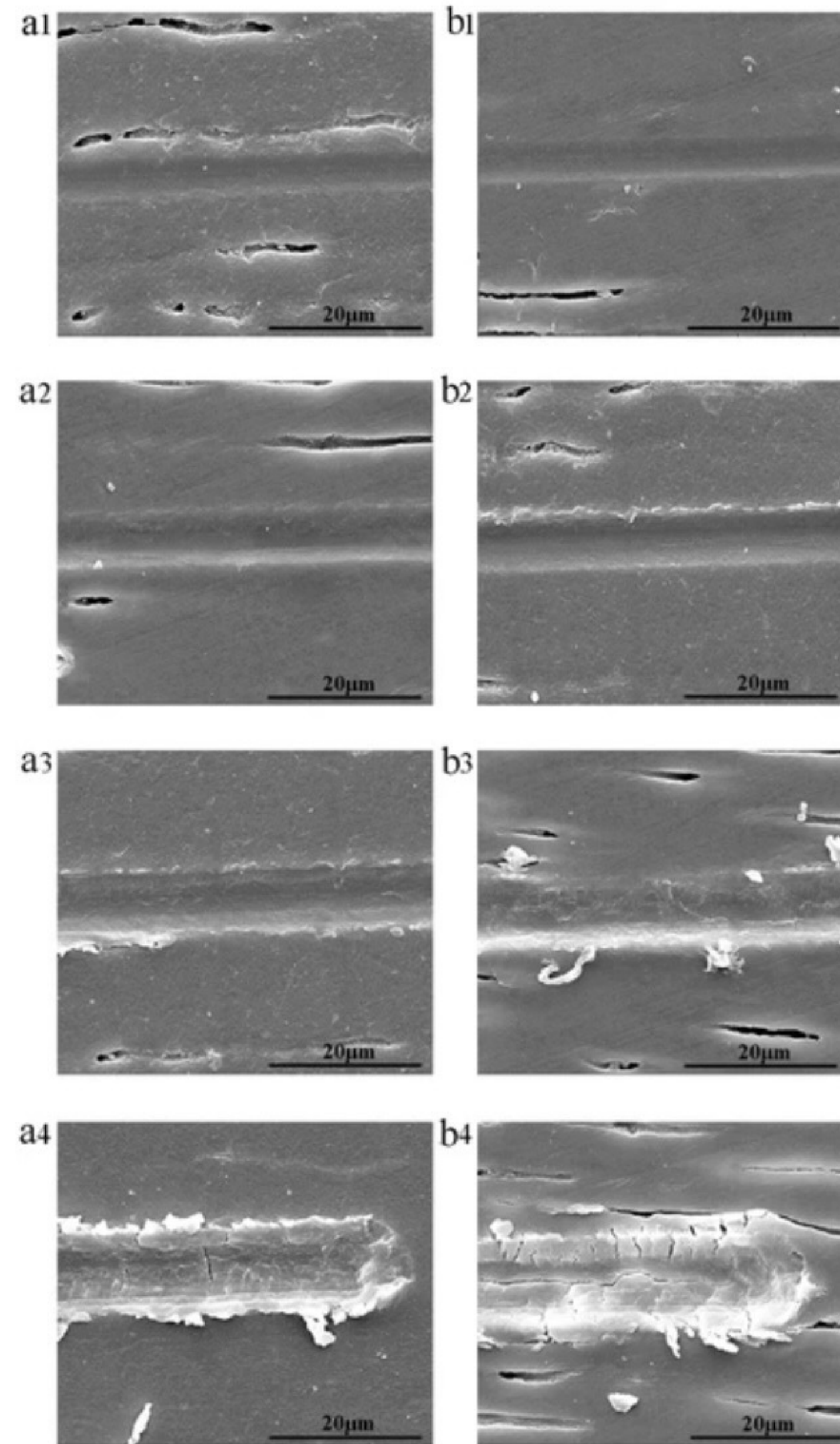
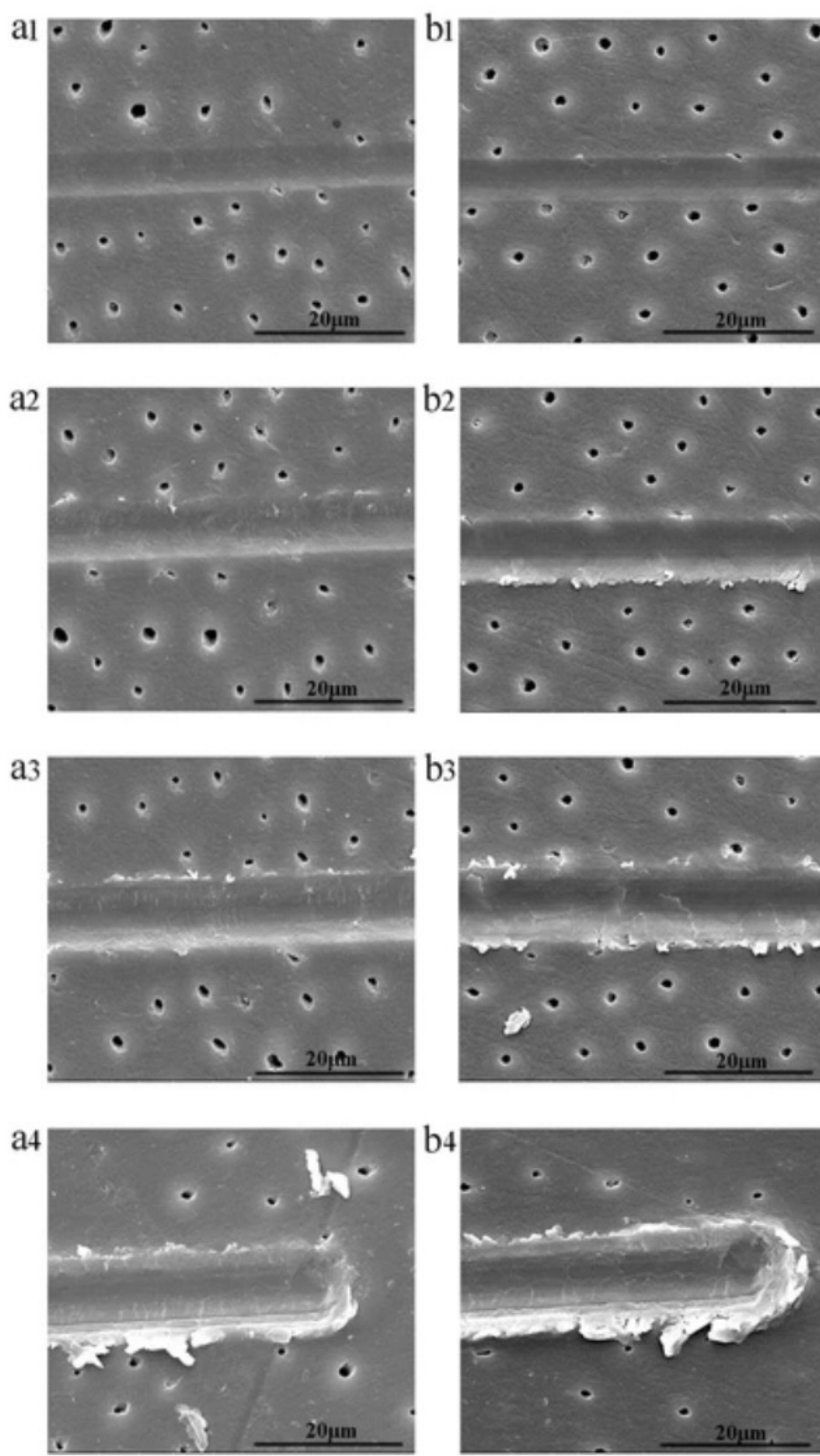
FTIR analysis

In both the dentin samples cut perpendicular to tubule orientation (Fig. 6a) as well as parallel to tubule orientation (Fig. 6b), the absorbance and the integrated area of CO_3^{2-} v2 and PO_4^{3-} v1, v3 significantly decreased after irradiation. In addition, the PO_4^{3-} v1, v3 integrated area reduced more than the CO_3^{2-} v2 area. The C/M ratio after irradiation increased significantly in both orientation groups (Table 2).

Fig. 6



Dentin samples cut perpendicular to tubule orientation as well as parallel to tubule orientation



Qing et al. reported that irradiation significantly altered the microwear behavior of dentin, showing increased debris, cracking, and delamination under higher loads (≥ 60 mN), which indicated reduced structural integrity and wear resistance compared to non-irradiated dentin.

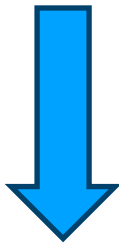
In summary:



- ❑ Irradiation **reduced crystallinity** and **phosphate content**, but **increased crystal size** and the **carbonate/mineral ratio** in dentin.
- ❑ Dentin became **softer** with **decreased microhardness** and **wear resistance**.
- ❑ **Cracks, tubular blockage**, and **collagen damage** appeared after irradiation.
- ❑ The dentin tissue became **dry, brittle, and weak**, especially at doses **≥60 Gy**.

Impact on Bonding

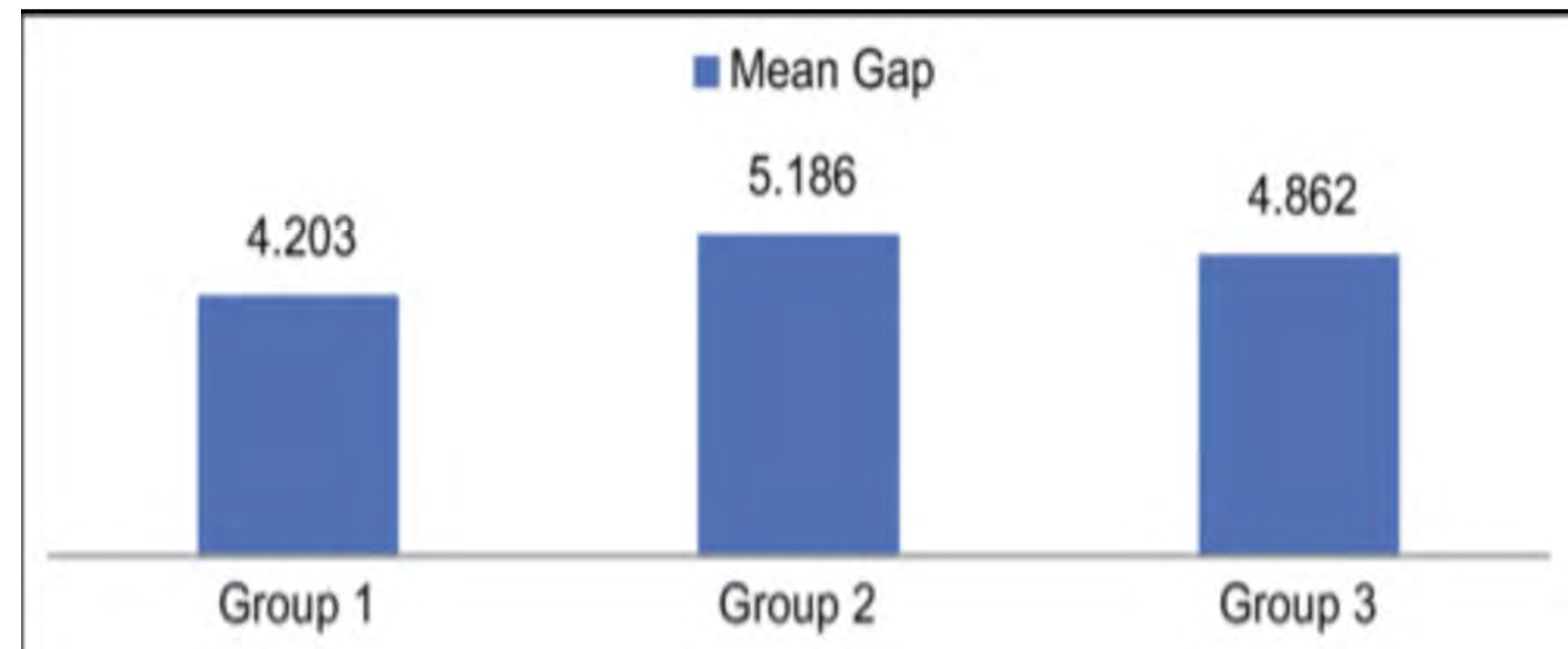
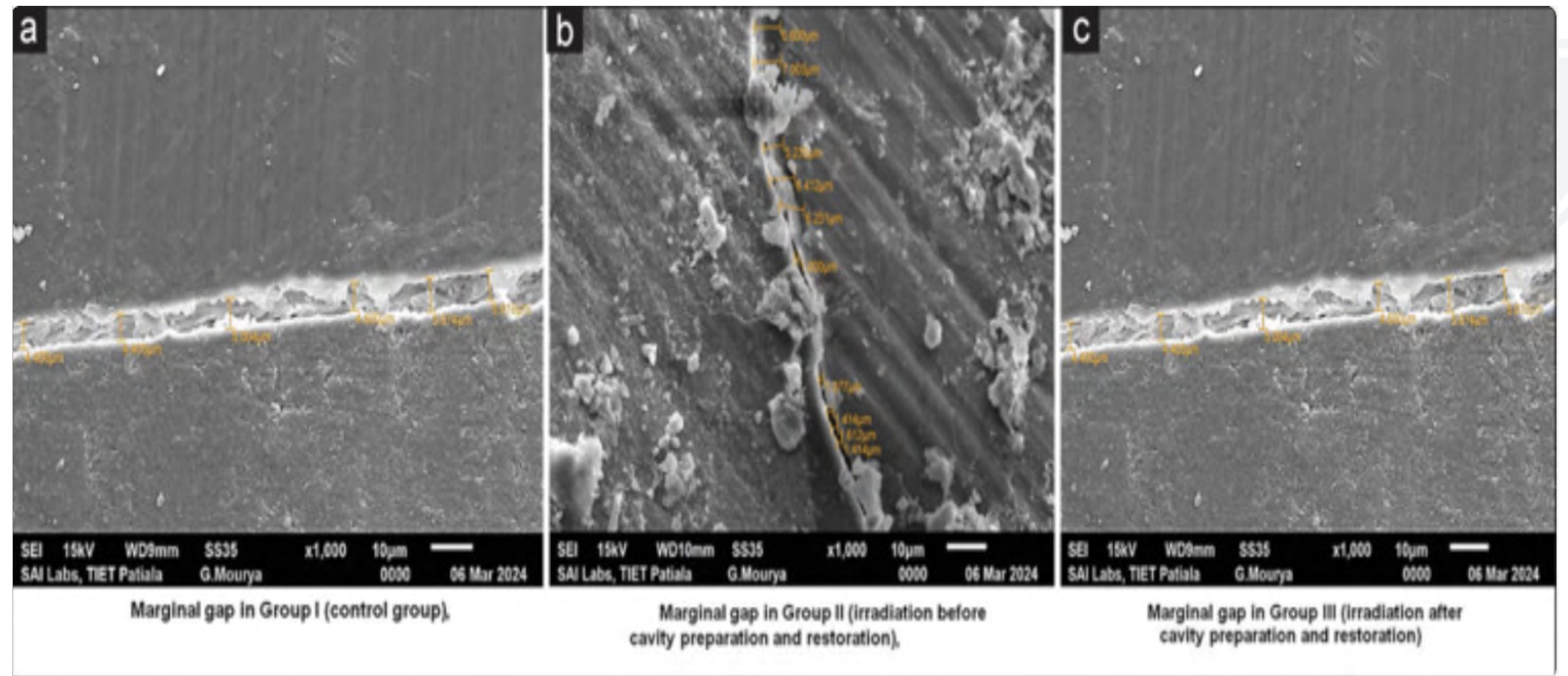


Bonding is done BEFORE radiation:	Bonding is done AFTER radiation:
Composite and adhesive have been polymerized  The pre-existing hybrid layer is not affected.  Strong bond strength, long-lasting restoration	Radiation alters the properties of dentin/enamel  Adhesive cannot form stable hybrid layer  Decreased bond strength, reduced restoration life-span

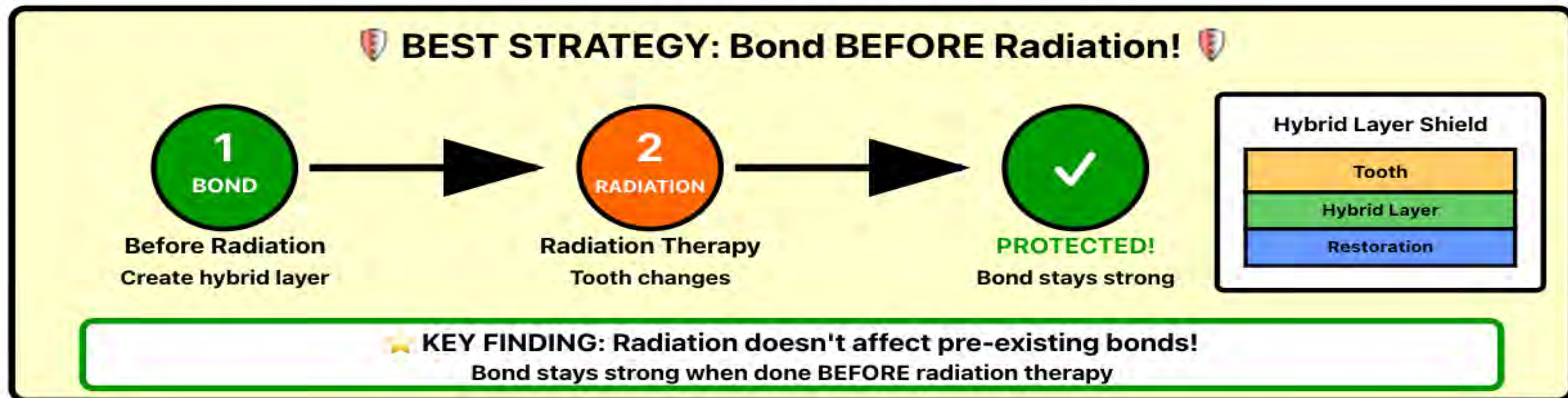
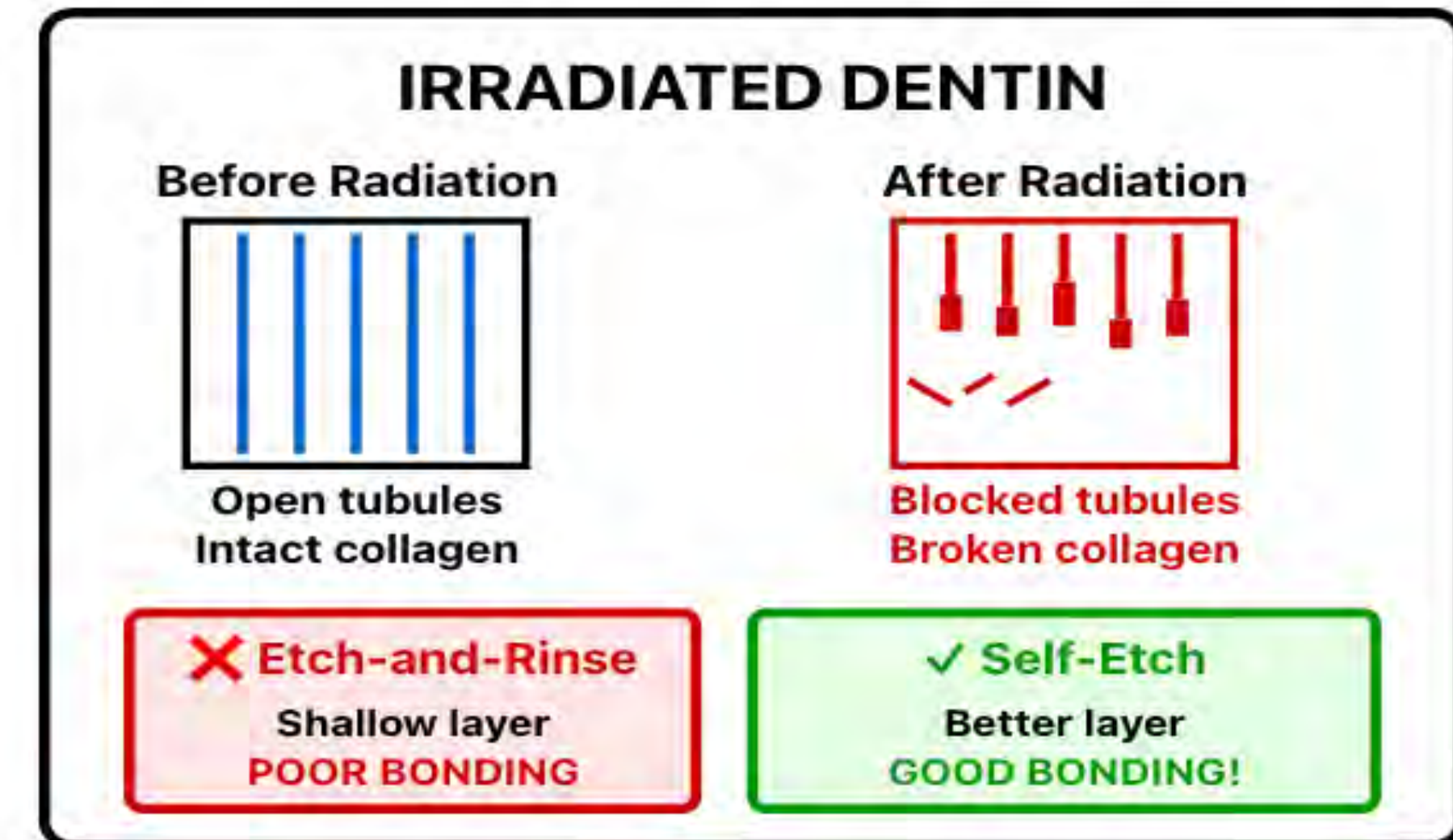
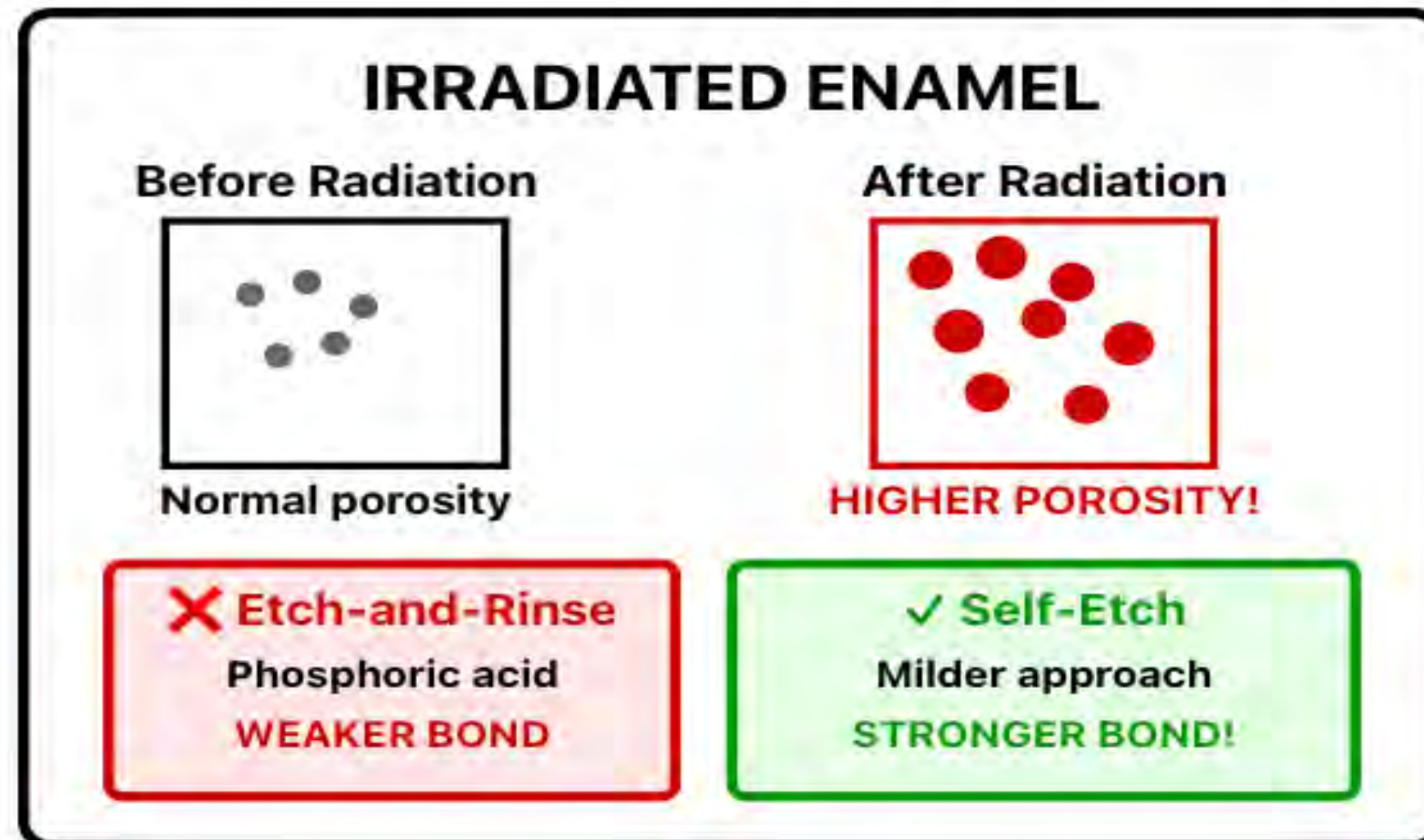
In conclusion: restorations should be done prior to receiving radiation to preserve the hybrid layer between adhesive systems and tooth structure. This will allow for longevity of the restoration.

Impact on Bonding

- It is recommended that restorative dental treatments should be completed prior to the initiation of radiotherapy to ensure optimal bonding and long-term restoration success.



Dental Bonding After Radiation Therapy



In Summary

- Radiation leads to **systemic effects** such as fatigue, weight loss, and immunosuppression.
- Several **oral effects**, the main one being:
 - Damage to salivary glands leads to:
 - Xerostomia
 - Radiation caries
- **Weakens enamel and dentin** wear resistance by altering its crystal structure, chemical composition, and surface microhardness.
- Impairs **bonding strength** of adhesive systems after radiation occurs.



Behavioral Objective 2

Differentiate between restorative materials (amalgam, composites, GIC, RMGIC, ceramics, zirconia, bioactive materials) by explaining how radiation may influence their clinical performance and interaction with oral tissues.

Aidan Rackliffe

Search Strategy

P: Teeth and restorative materials exposed to head and neck radiotherapy

I: Ionizing radiation (therapeutic radiation doses **30–80 Gy**)

C: Non-irradiated control materials or pre-radiation restorations

O: Changes in mechanical, physical, and bonding **properties**; clinical **performance** and longevity

Databases and Searches used



Keywords

restorative materials, head and neck radiation, bond strength

Pubmed search string

((Restorative materials) AND (head and neck radiation)) OR "ionizing radiation"("restorative materials" OR composite OR amalgam OR "glass ionomer" OR "RMGIC" OR ceramic OR zirconia OR bioactive)

Search results

120 results
12 selected
6 used

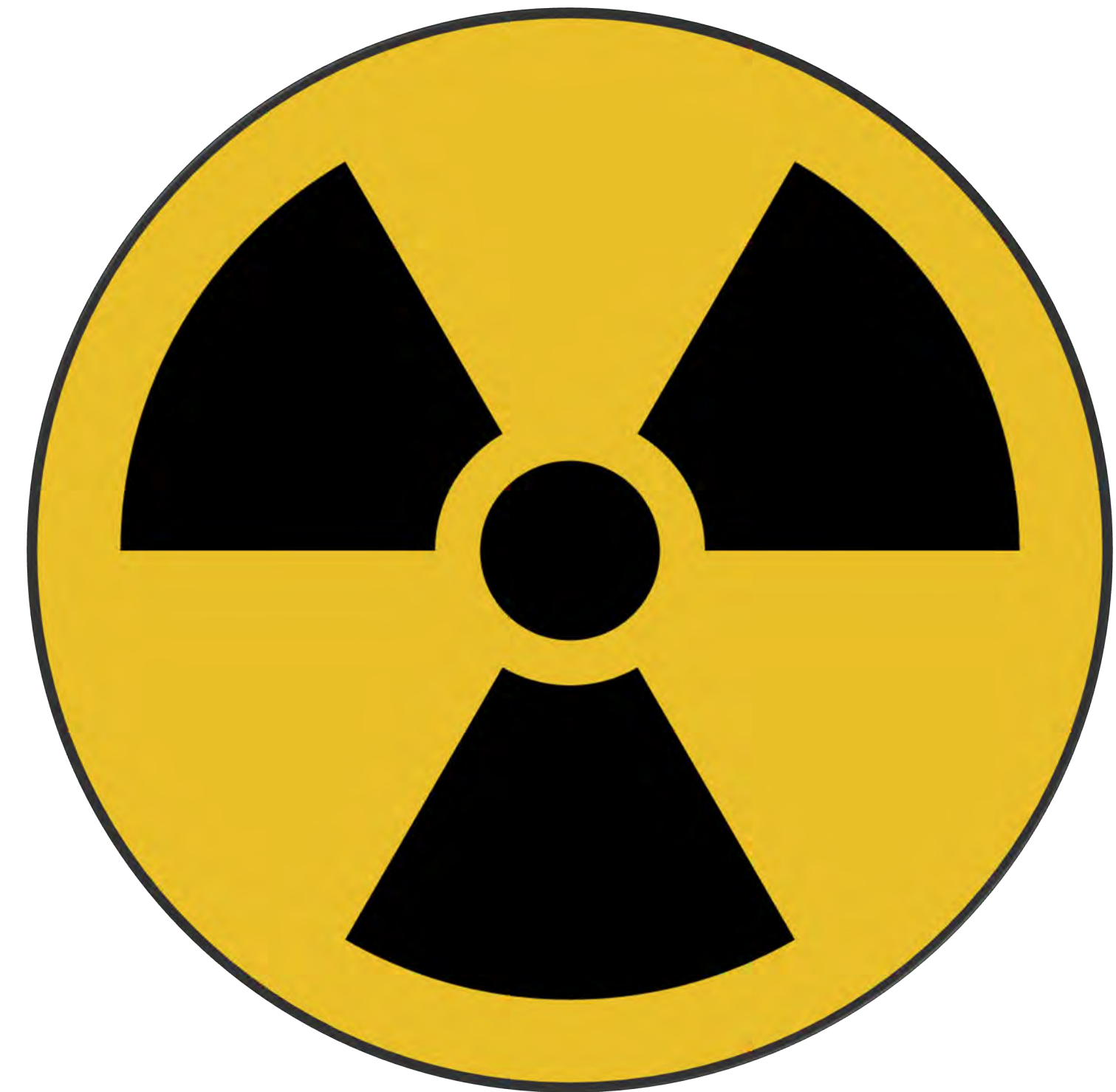
Highest level of evidence

Systematic reviews, systematic reviews of randomized clinical trials

Why Focus on Materials?



- Ionizing radiation affects enamel, dentin, and restorative interfaces.
- Strength and bonding vary by material composition.
- Goal: **Preserve tooth structure and select materials that maintain performance under RT.**



Amalgam and Metallic Restorations



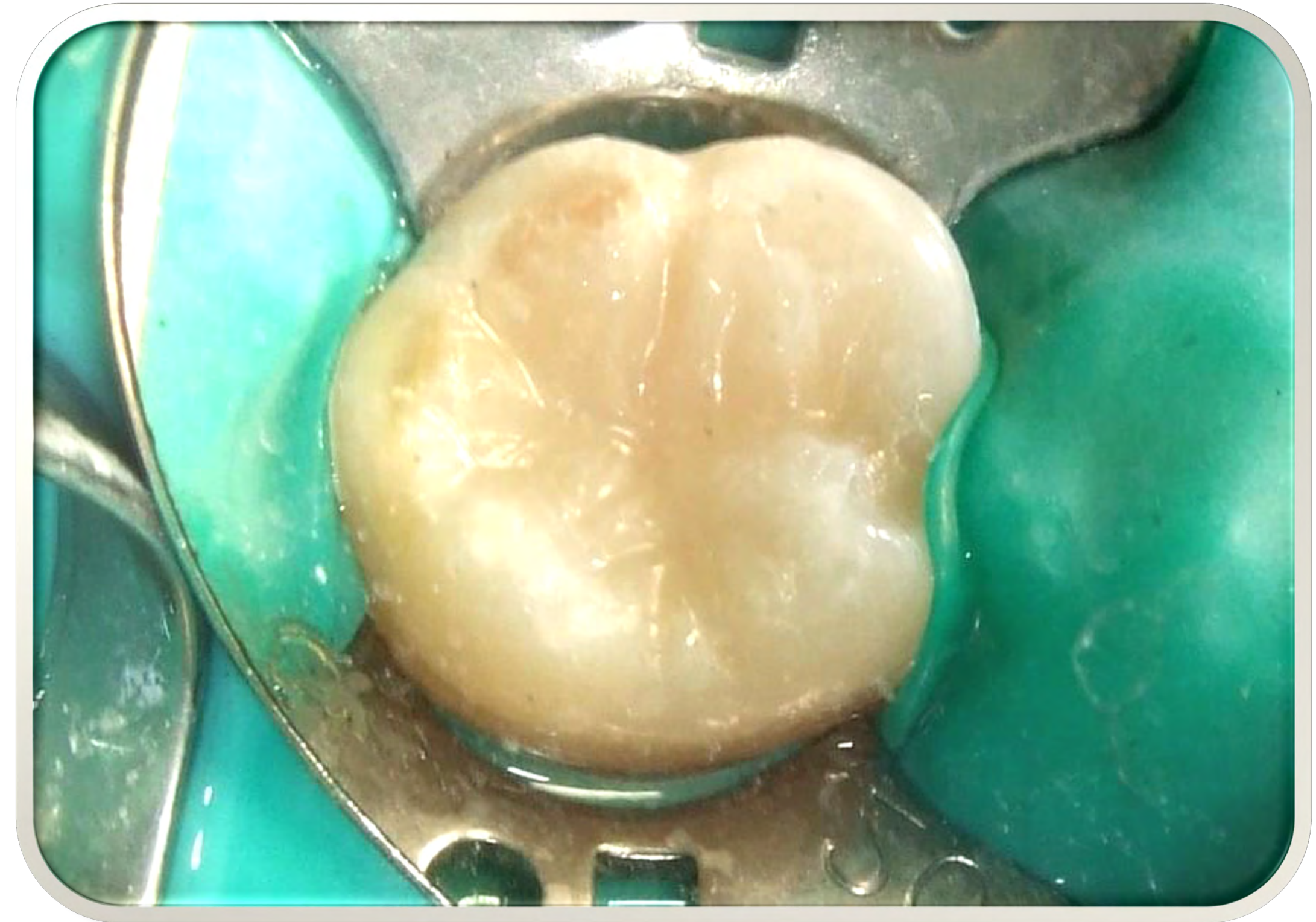
- **After RT:** Stable structure but amplifies photon energy → ↑ backscatter (Oktaý 2022).
- **Fluoride/Ion Release:** None.
- **Xerostomia:** Neutral — no harm from dryness, but not ideal due to prep weakening irradiated teeth.
- **Evidence:** Multiple in vitro dosimetry studies confirm ~10–35% localized dose enhancement



Resin Composites



- **After RT:** Mechanically stable but post-RT placement weakens bonding (collagen oxidation)
- **Fluoride/Ion Release:** None
- **Xerostomia:** Performs well with fluoride varnish maintenance.
- **Evidence:** Consistent findings from Madrid Troconis (2017) and Eggmann (2023).



Conventional GLC

- **After RT:** Hardness and fluoride release may change; microcracks form (Amorim 2021).
- **Fluoride/Ion Release:** Fluoride release increases transiently, promoting remineralization.
- **Xerostomia:** Loses surface integrity faster; not durable long term.
- **Evidence:** Sivavong (2024) – GLC weakest post-RT performance.



Resin Modified Glass Ionomer



- **After RT:** Maintains hardness and shows higher compressive strength than GIC (Sivavong 2024).
- **Fluoride/Ion Release:** Sustained release; resin network aids resistance.
- **Xerostomia:** Still prone to surface softening but more resilient than GIC.
- **Evidence:** Supported Sivavong 2024 and Cekalovic 2024 studies.



Ceramics and Zirconia

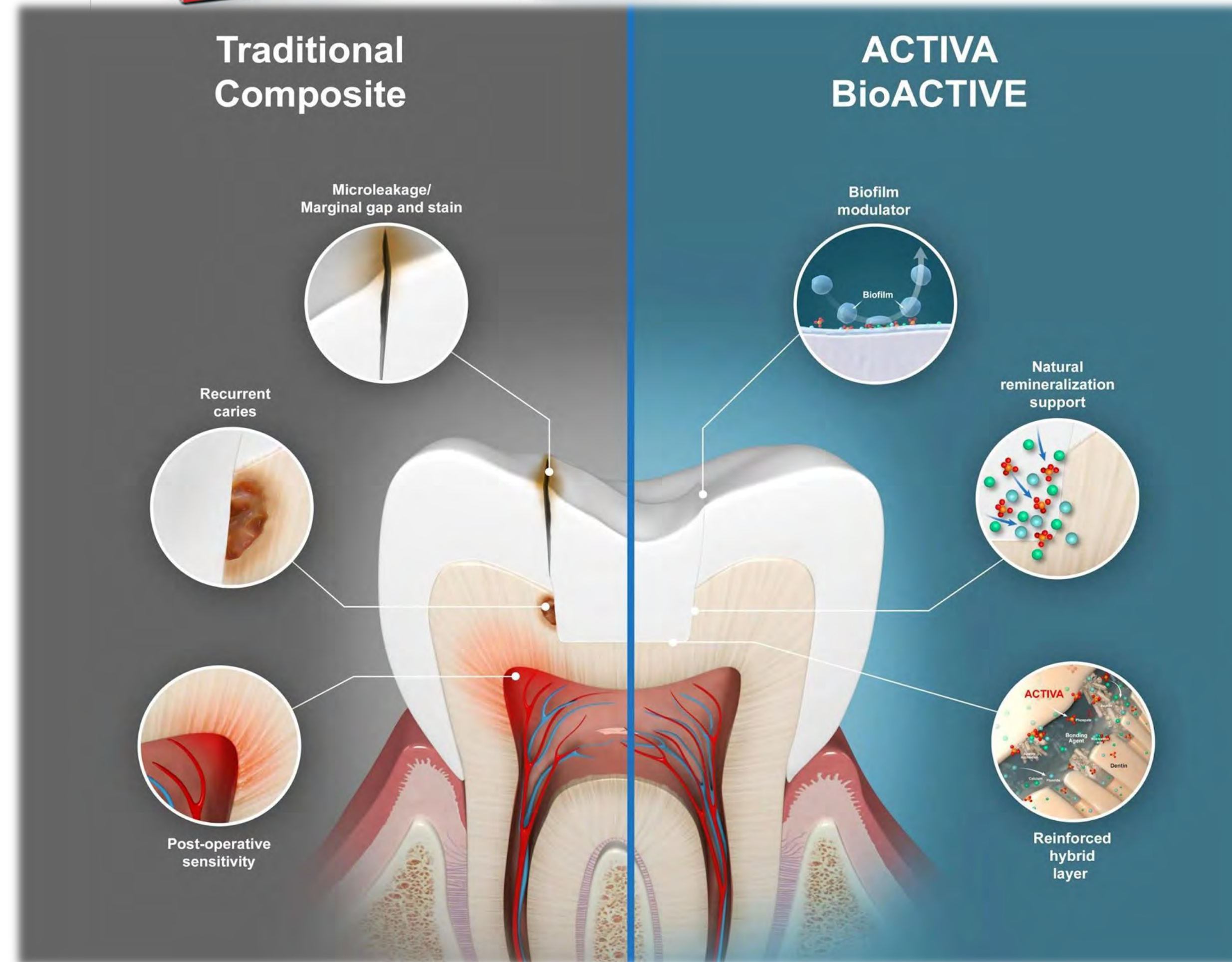


- **After RT:** Chemically and dimensionally stable; no degradation.
- **Fluoride/Ion Release:** None
- **Xerostomia:** Neutral — no harm from dryness, but not ideal due to prep weakening irradiated teeth.
- **Evidence:** Moderate: good radiation data, limited clinical outcomes post-RT and possibility of backscatter.



Bioactive Materials

- **After RT:** Retain strength and show favorable surface integrity (Atalay 2024).
- **Fluoride/Ion Release:** Continuous release of Ca^{2+} , PO_4^{3-} , and F^- ions (Sivavong 2024).
- **Xerostomia:** Excellent potential—enhances remineralization and pH buffering.
- **Evidence:** Promising but limited; more clinical trials needed.



Material	What Happens After RT?	Fluoride / Ion Release	Okay in Xerostomia?
Amalgam / Metals	Structurally stable but can increase local dose via backscatter (10–35%).	None	Neutral – safe but invasive prep weakens teeth.
Composite Resins	Mechanically stable but post-RT dentin bonding weakens due to collagen oxidation.	None	Good – strong seal and long-term stability.
Conventional GIC	Decreased hardness and surface integrity after RT.	Moderate, short-term fluoride release.	Fair – helps remineralization but less durable.
RMGIC	Maintains hardness and compressive strength post-RT.	Sustained fluoride release.	Fair – more resilient than GIC.
Ceramics / Zirconia	Chemically and mechanically stable after RT.	None	Neutral – safe but invasive prep weakens teeth.
Bioactive Materials	Maintain strength and surface integrity post-RT.	Continuous Ca^{2+} , PO_4^{3-} , F^- ion release.	Good – promote remineralization and pH balance.



Compare and contrast the advantages and limitations of restorative material choices based on scientific evidence in the context of radiation-induced changes such as xerostomia, tooth brittleness, and altered bonding.

Daniel Lee

Search Strategy

P: Patients with **radiation-induced** oral changes (xerostomia, brittle teeth, altered bonding)

I: Use of different dental restoration choices (GIC, RMGIC, Resin Composite)

C: Comparing different restoration choices ((GIC, RMGIC, Resin Composite)

O: Improved bonding durability, resistance to brittleness, and longevity in irradiated teeth

Databases used

Pubmed

Keywords

Restorative materials (RMGI, GI, Composite), Xerostomia, Brittleness, Bonding

Pubmed search string

((Restorative materials) AND (Xerostomia OR brittleness OR bonding) AND (radiation therapy))

Search results

95 results
9 selected
5 used

Highest level of evidence

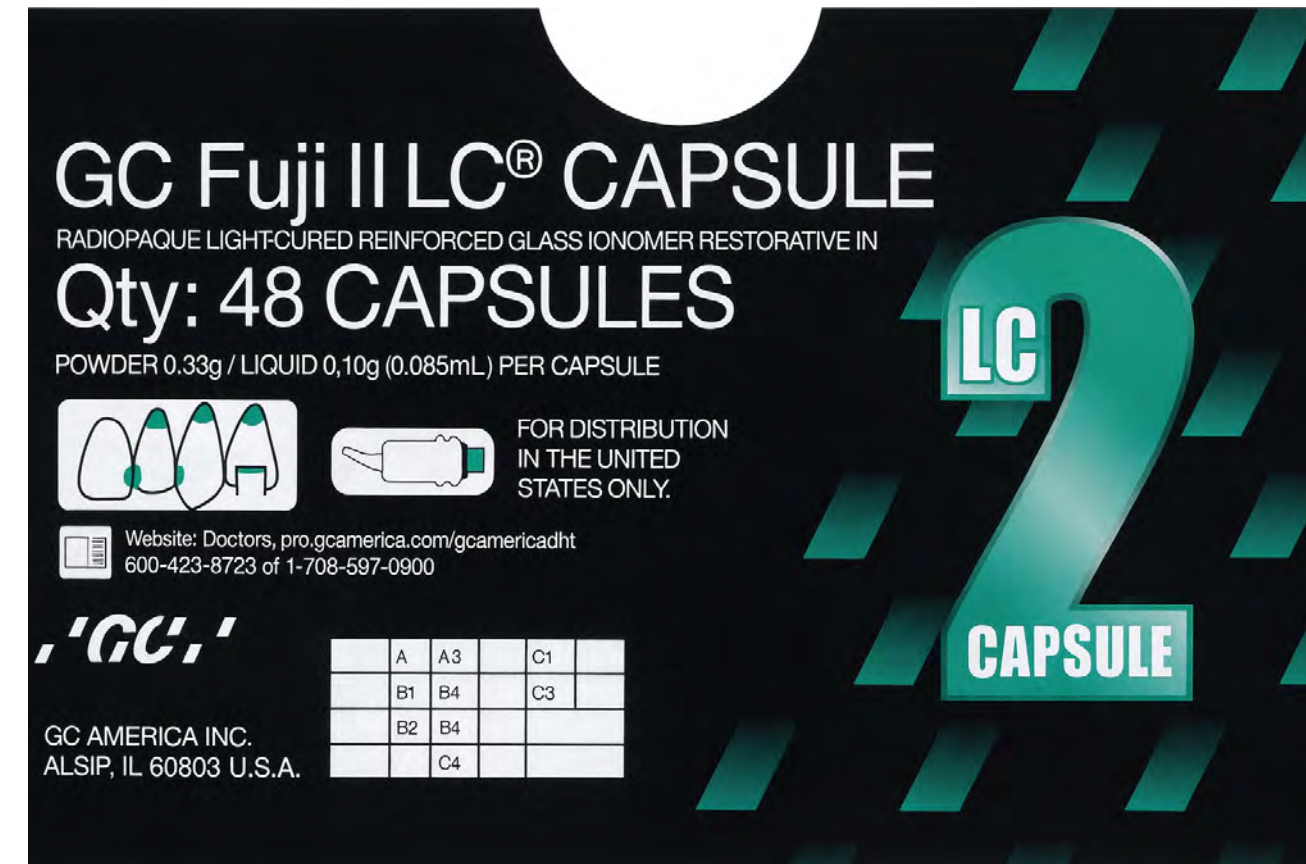
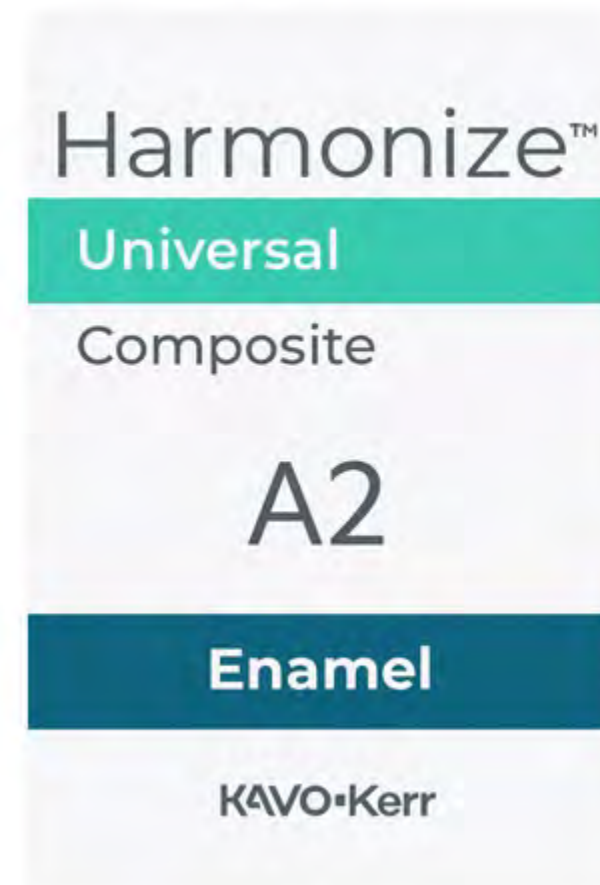
Systematic reviews, systematic reviews of randomized clinical trials



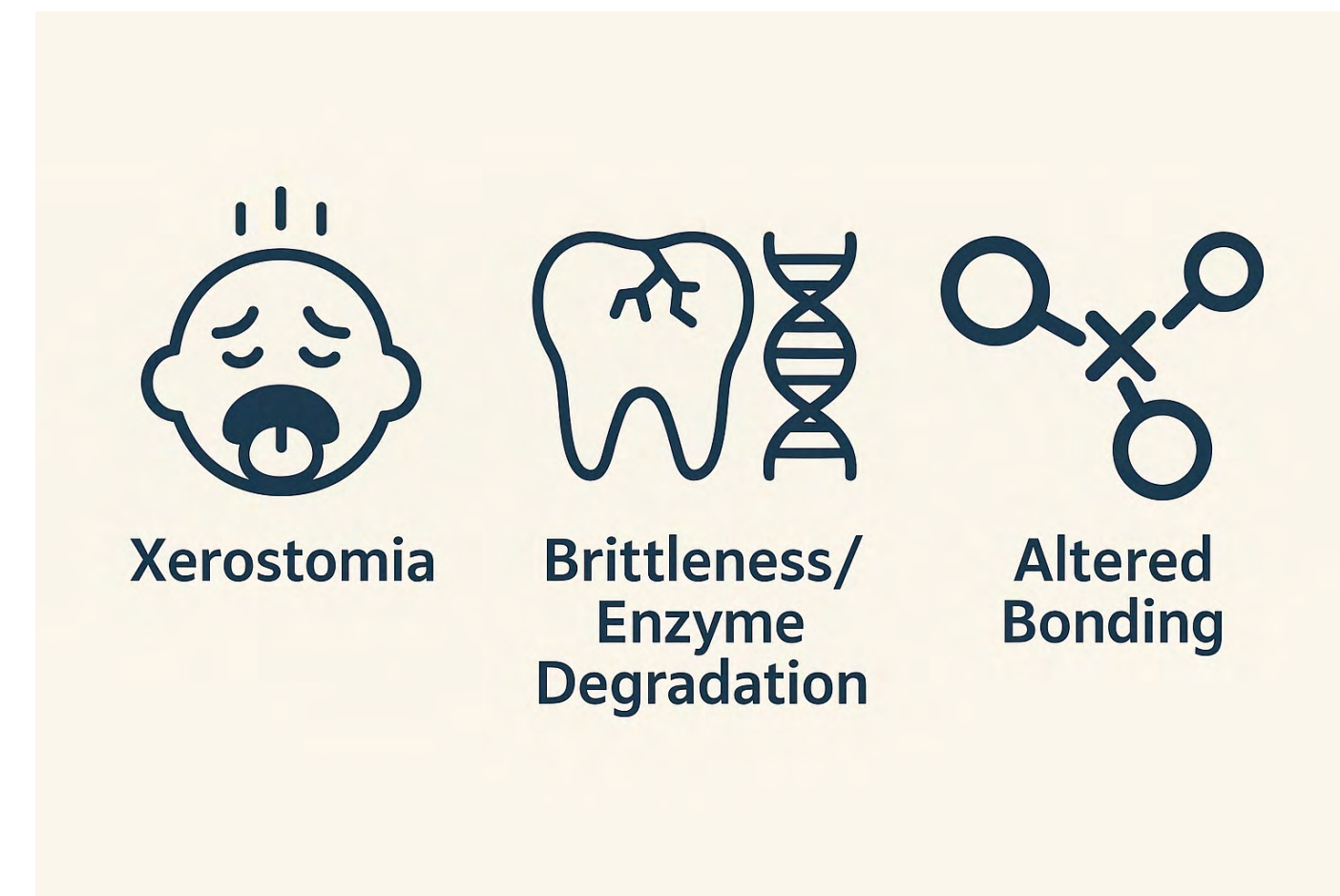
Materials and Conditions



- Material Choice
 - Composite Resin
 - RMGI
 - GI



- Conditions
 - Xerostomia
 - Brittleness/Enzyme Degradation
 - Altered Bonding



Fluoride Release



► J Dent (Tehran). 2013 Mar 31;10(2):134–140.

Comparison of the Amount of Fluoride Release from Nanofilled Resin Modified Glass Ionomer, Conventional and Resin Modified Glass Ionomer Cements

Sumitha Upadhyay¹, Arathi Rao^{2,✉}, Ramya Shenoy³

► Author information ► Article notes ► Copyright and License information

PMCID: PMC3666073 PMID: [23724212](#)

- In vitro study
- Fluoride release measured in water bath
- RMGI releases less Fluoride until 15 days after procedure

Time Period (days)	Conventional GIC (ppm)	RMGIC (ppm)	Composite Resin
1	3.38 ± 0.42	2.88 ± 0.49	Minimal/None
2	2.08 ± 0.18	0.77 ± 0.19	Minimal/None
7	2.51 ± 0.31	2.06 ± 0.18	Minimal/None
15	1.42 ± 0.16	2.00 ± 0.14	Minimal/None
30	1.08 ± 0.15	2.19 ± 0.15	Minimal/None

Bond Strength to Dentin after 24 Hours



In Vitro Comparison of Surface Roughness, Flexural, and Microtensile Strength of Various Glass-Ionomer-Based Materials and a New Alkasite Restorative Material

Alper Kaptan ¹, Fatih Oznurhan ², Merve Candan ³

Affiliations + expand

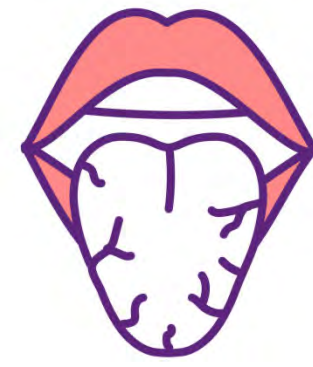
PMID: 36771950 PMCID: PMC9920171 DOI: 10.3390/polym15030650

Table 3. The microtensile bond strength (μ TBS) values for each dental restorative material with teeth.

Materials	Mean $\pm$ Standard Deviation (MPa)	
Group 1 (LC-CENT)	13.25 $\pm$ 4.4 ^B	
Group 2 (SC-CENT)	12.50 $\pm$ 5.05 ^B	
Group 3 (COMP)	16.50 $\pm$ 7.73 ^A	
Group 4 (DYRA)	16.36 $\pm$ 4.64 ^A	F = 2.873 p = 0.015 *
Group 5 (LINER)	11.98 $\pm$ 5.01 ^B	
Group 6 (FUJI)	11.17 $\pm$ 3.0 ^B	
Group 7 (NOVA)	9.88 $\pm$ 1.87 ^C	

The different uppercase letters represent the difference in the columns. * $p < 0.05$ was accepted as the significance level.

- Group 3 – Resin Composite
- Group 5 and 6 – RMGI
- Group 7 – GI
- "Many factors affect the results, such as tooth age, depth of dentine, dentine tubules, moisture, materials, and test conditions."



Xerostomia



Meta-Analysis > J Prosthet Dent. 2022 Nov;128(5):886-896. doi: 10.1016/j.prosdent.2021.02.002.
Epub 2021 Mar 11.

Impact of head and neck radiotherapy on the longevity of dental adhesive restorations: A systematic review and meta-analysis

Natália Rangel Palmier ¹, Cristhian Camilo Madrid Troconis ², Ana Gabriela Costa Normando ³,
Eliete Neves Silva Guerra ⁴, Anna Luíza Damaceno Araújo ³, Lady Paola Aristizábal Arboleda ³,
Jéssica Montenegro Fonsêca ³, Mariana de Pauli Paglioni ⁵, Wagner Gomes-Silva ⁶,
Aljomar José Vechiato Filho ⁶, Wilfredo Alejandro González-Arriagada ⁷,
Adriana Franco Paes Leme ⁸, Ana Carolina Prado-Ribeiro ⁶, Thaís Bianca Brandão ⁹,
Mario Fernando de Goes ¹⁰, Marcio Ajudarte Lopes ¹¹, Alan Roger Santos-Silva ¹²

Composite

- does not release fluoride
- best structural integrity in low moisture

GIC

- high initial release of fluoride, tapering off
- structure damaged in xerostomic patients due to low moisture



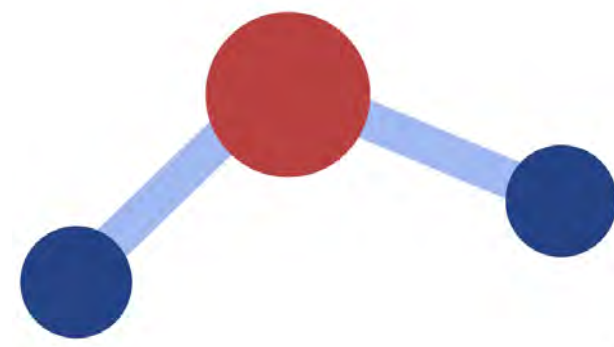
More Ideal

RMGI

- continuous, moderate release of fluoride
- intermediate structure

Less Ideal





Altered Bonding



Meta-Analysis > J Prosthet Dent. 2022 Nov;128(5):886-896. doi: 10.1016/j.prosdent.2021.02.002.
Epub 2021 Mar 11.

Impact of head and neck radiotherapy on the longevity of dental adhesive restorations: A systematic review and meta-analysis

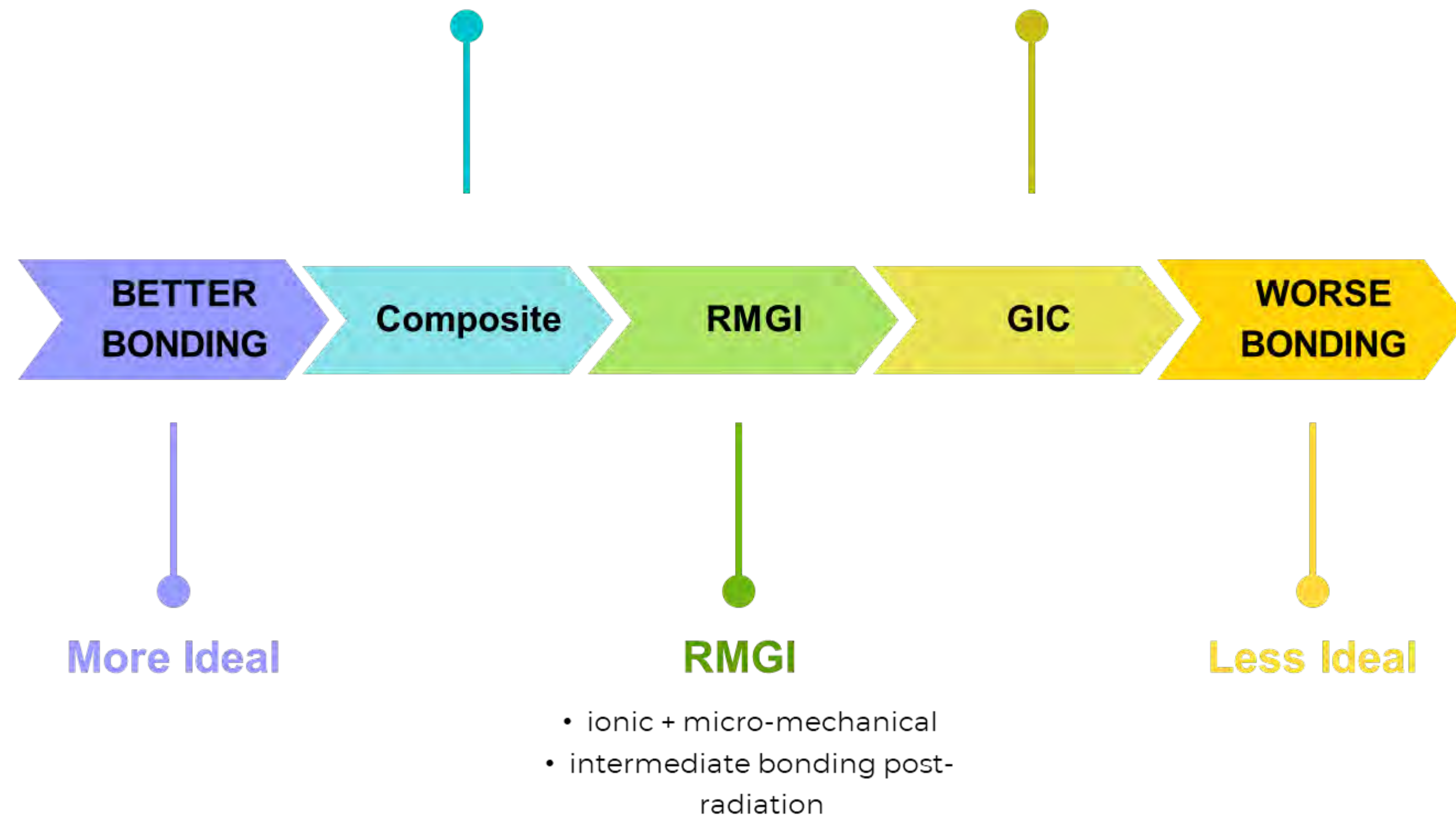
Natália Rangel Palmier ¹, Cristhian Camilo Madrid Troconis ², Ana Gabriela Costa Normando ³,
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Adriana Franco Paes Leme ⁸, Ana Carolina Prado-Ribeiro ⁶, Thaís Bianca Brandão ⁹,
Mario Fernando de Goes ¹⁰, Marcio Ajudarte Lopes ¹¹, Alan Roger Santos-Silva ¹²

Composite

- micro-mechanical
- collagen fibers less affected than chemical bonding

GIC

- Ionic bonding worsens post-radiation
- reduced calcium availability





Brittleness - Hydrolytic/Enzymatic Degradation

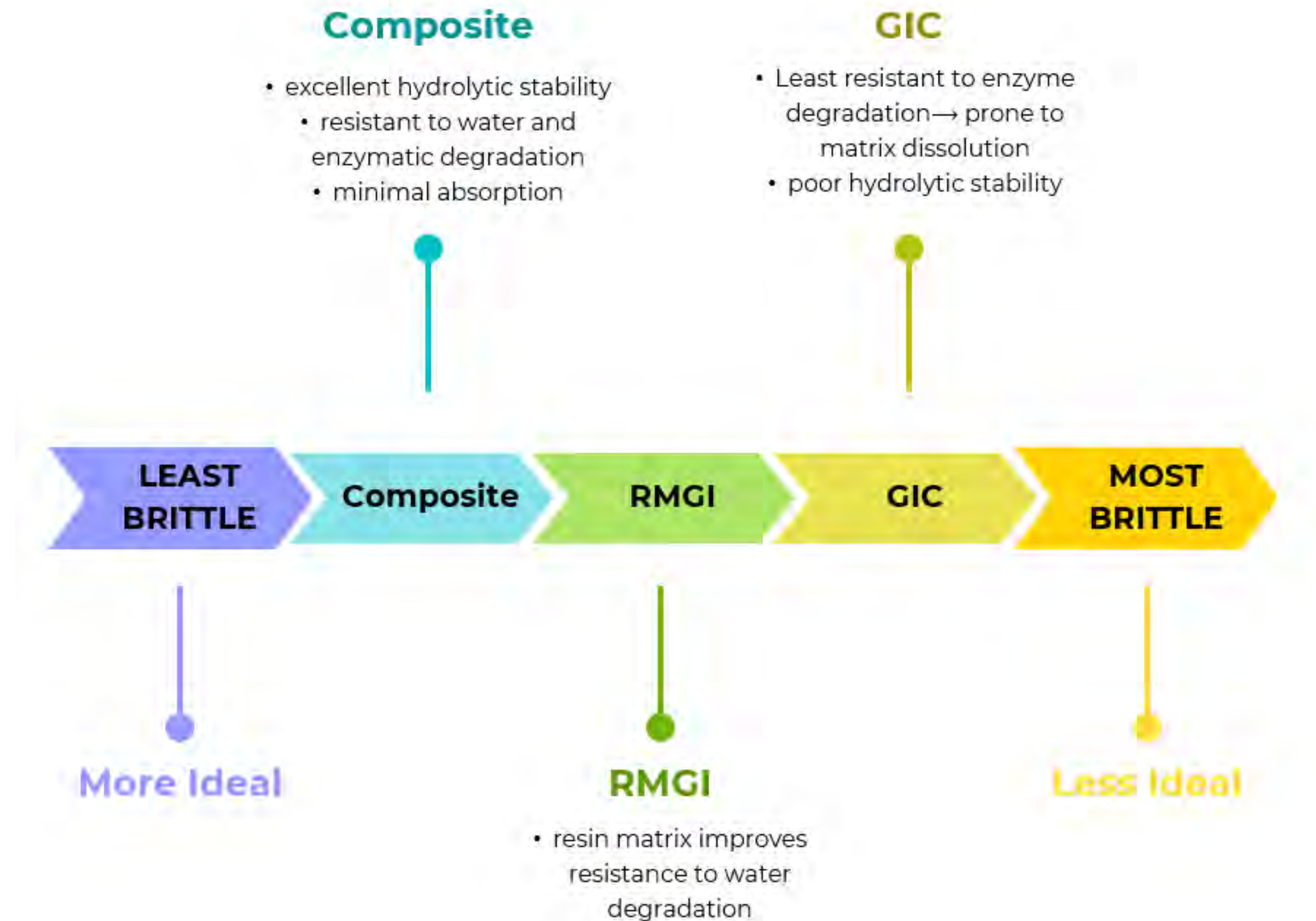


Meta-Analysis > J Prosthet Dent. 2022 Nov;128(5):886-896. doi: 10.1016/j.prosdent.2021.02.002.
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Impact of head and neck radiotherapy on the longevity of dental adhesive restorations: A systematic review and meta-analysis

Natália Rangel Palmier¹, Cristhian Camilo Madrid Troconis², Ana Gabriela Costa Normando³, Eliete Neves Silva Guerra⁴, Anna Luíza Damaceno Araújo³, Lady Paola Aristizábal Arboleda³, Jéssica Montenegro Fonsêca³, Mariana de Pauli Paglioni⁵, Wagner Gomes-Silva⁶, Aljomar José Vechiato Filho⁶, Wilfredo Alejandro González-Arriagada⁷, Adriana Franco Paes Leme⁸, Ana Carolina Prado-Ribeiro⁶, Thaís Bianca Brandão⁹, Mario Fernando de Goes¹⁰, Marcio Ajudarte Lopes¹¹, Alan Roger Santos-Silva¹²

- Radiation therapy increases activity of MMP (matrix metalloproteinase)



Composite Bonding

- **Etch and Rinse**
 - Over-etched enamel and dentin
 - Already porous from radiation therapy
 - Heavily dependent on collagen bonding
- **Self-Etch**
 - Milder acidic monomers
 - Less dependent on collagen bonding
 - Hydroxyapatite bonding

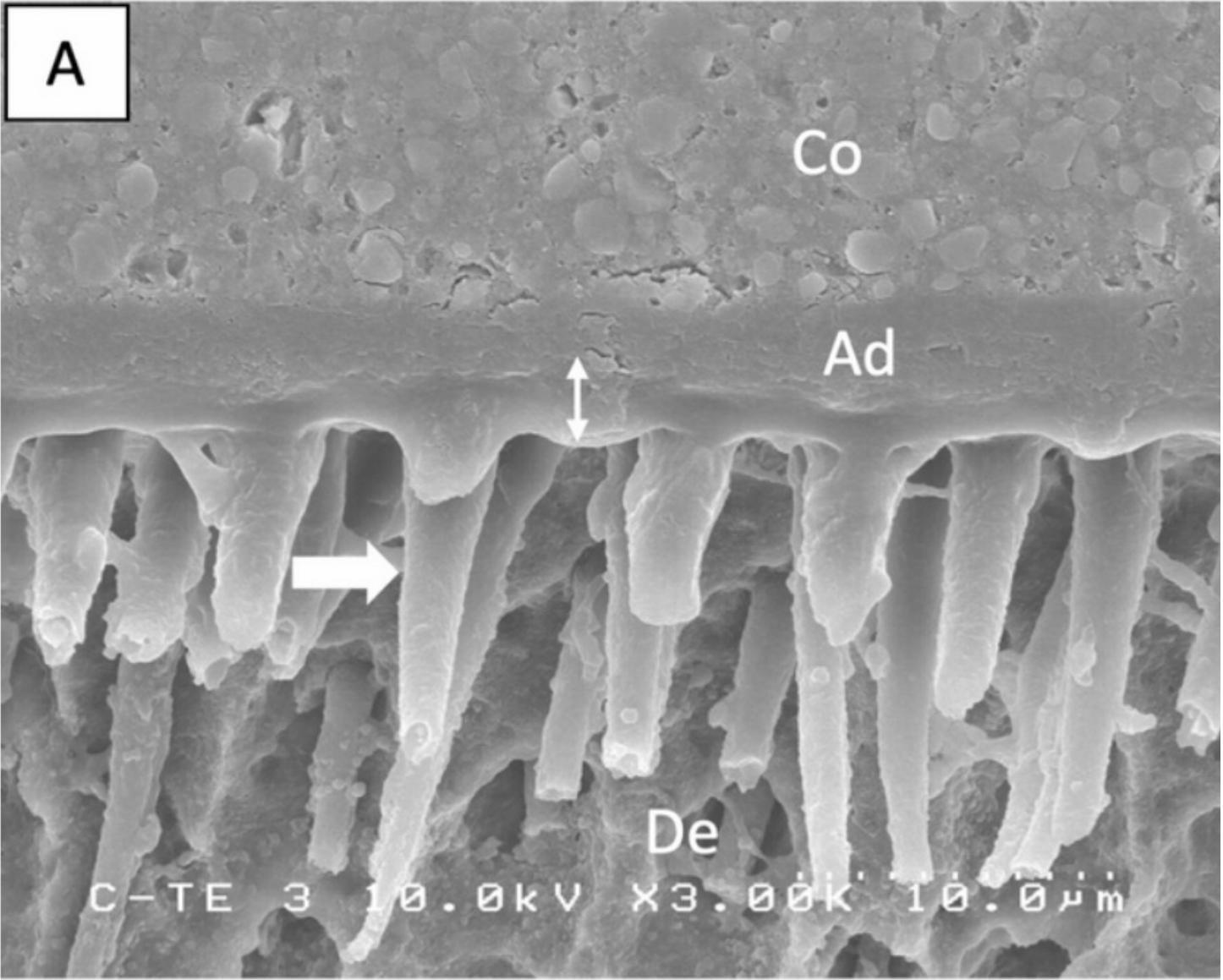


Effect of radiation therapy on the dentin bond strength of a universal adhesive

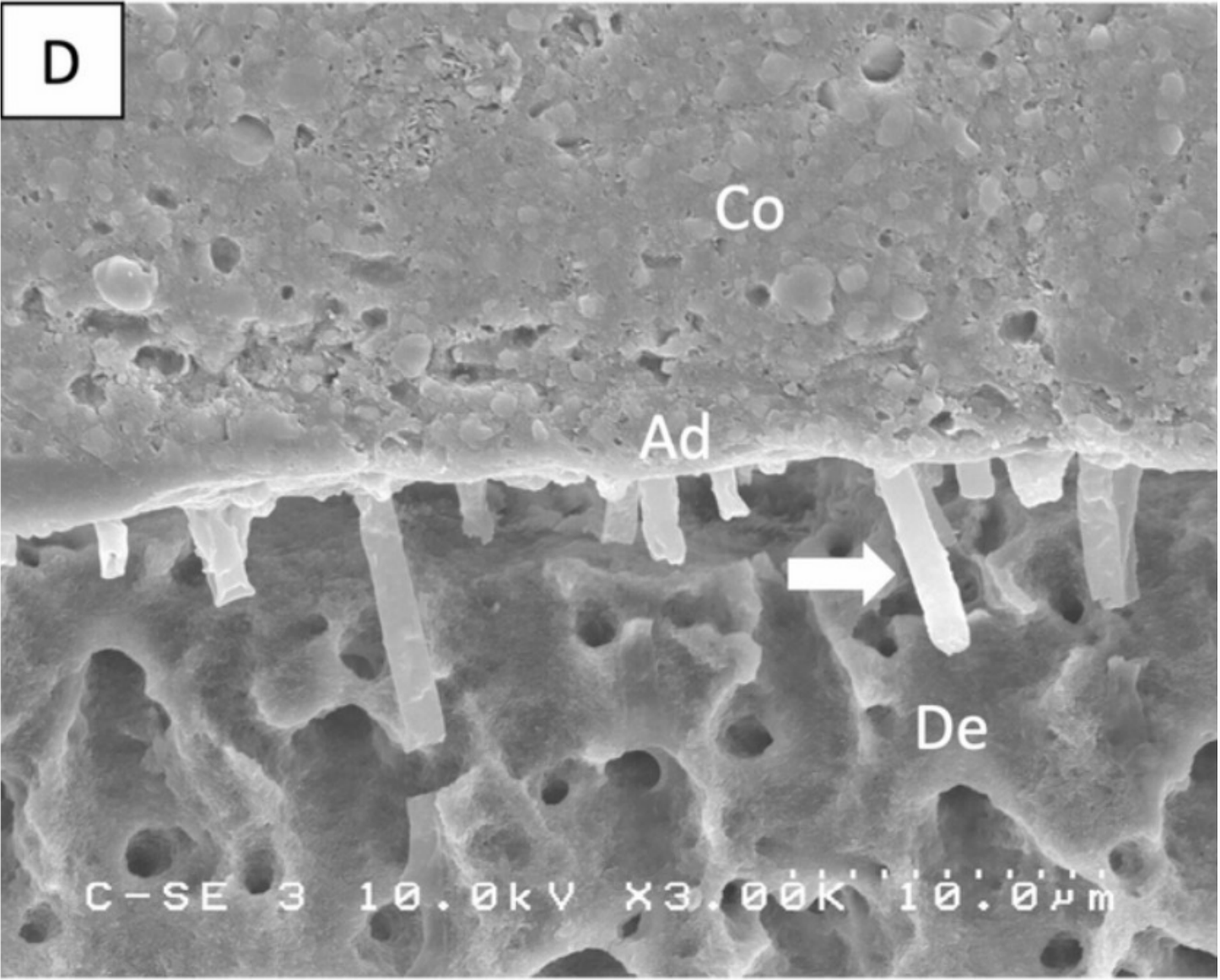
Pipop Saikaew ¹, P Hannongbua ², P Jianratanajit ², I Wongkertprayot ², P Chaiklahan ², N Srimaneekarn ³, A Prayongrat ⁴, P Eamsa-Ard ⁵, Afma Chowdhury ⁶, K Katekovit ², S Hidehiko ⁷

Affiliations + expand
PMID: 40426161 PMCID: PMC12117943 DOI: 10.1186/s12903-025-06226-5

Etch-and-Rinse



Self-etch



Radiotherapy impairs adhesive bonding in permanent teeth

Juliana Arid ¹, Regina Guenka Palma-Dibb ², Harley Francisco de Oliveira ³, Paulo Nelson-Filho ⁴,
Fabricio Kitazono de Carvalho ⁴, Lea Assed Bezerra da Silva ⁴, Talitha de Siqueira Mellara ⁴,
Raquel Assed Bezerra da Silva ⁴, Juliana Jendiroba Faraoni ², Alexandra Mussolino de Queiroz ⁴

Affiliations + expand

PMID: 31020438 DOI: 10.1007/s00520-019-04782-5



	SB	CL
G1	20.99 ± 6.55 a B	18.95 ± 3.58 a B
G2	7.98 ± 4.24 a A	12.19 ± 6.18 b C
G3	14.93 ± 6.15 a AB	27.06 ± 8.05 a A
G4	15.46 ± 8.16 a B	22.03 ± 8.78 b B

- G1: Control (no radiation)
- G2: Restoration BEFORE radiation
- G3: Restoration immediately AFTER radiation
- G4: Restoration 6 months after radiation

- Adper Single Bond 2** (etch-and-rinse, SB)
- Clearfil SE Bond** (self-etch, CL)





Radiation therapy AFTER Restoration

Radiation therapy BEFORE Restoration

Application mode	Microtensile Bond Strength (MPa)			
	Control	F-RT	RT-F	
Etch-and-rinse	48.24 ± 3.50 ^A	48.71 ± 11.00 ^A	34.74 ± 9.56 ^B	
Self-etch	49.14 ± 7.77 ^A	48.80 ± 10.52 ^A	37.31 ± 4.57 ^B	

Radiation therapy does **not significantly
affect the hybrid layer that was formed
before irradiation**



Bioactive Composites



Time/Radiation Group - Subgroup	24 h/NI	24 h/ID	30 d/NI	30 d/ID
CC	20.1 (7.9)Aab	11.0 (2.7)Ac	22.2 (4.7)Aa	14.3 (3.1)Abc
BCI	17.1 (5.3)Aab	13.7 (3.6)Ab	21.3 (5.1)Aa	16.6 (4.9)Aab
BCII	17.8 (6.4)Aa	11.6 (3.2)Aa	17.1 (5.3)Aa	15.9 (5.4)Aa
BCIII	19.6 (4.6)Aab	13.4 (3.4)Ab	20.2 (5.6)Aa	16.6 (5.5)Aab

- Newer material
- Potentially better bonding post radiation therapy
 - **CC**: bond strength ↓ significantly after irradiation
 - **BC I–III**: no significant reduction
- Need more research



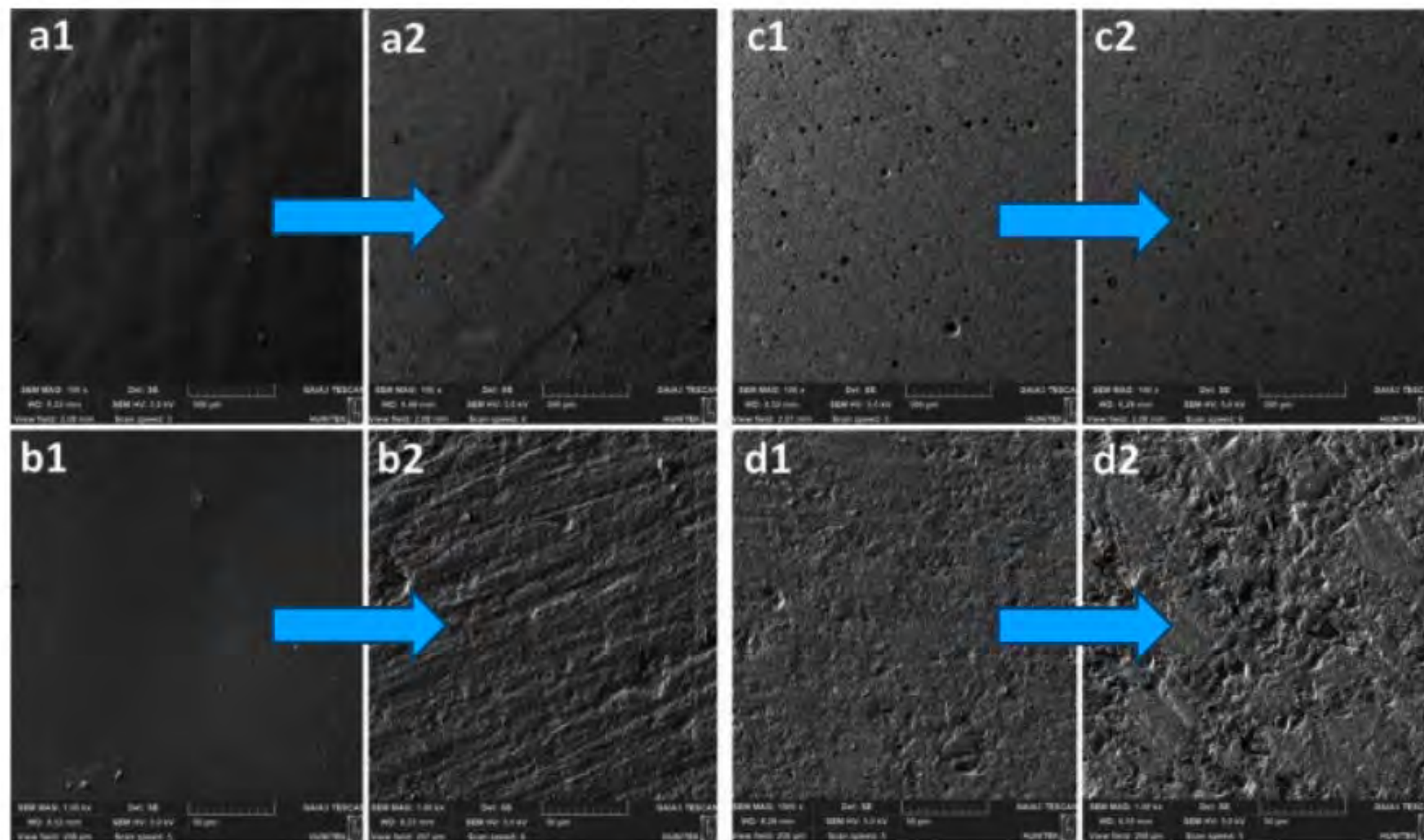


Fig 1. Conventional Resin Composite

► Support Care Cancer. 2024 Apr 18;32(5):295. doi: [10.1007/s00520-024-08476-5](https://doi.org/10.1007/s00520-024-08476-5)

Effect of radiotherapy on the surface roughness and microhardness of contemporary bioactive restorative materials

[Cansu Atalay](#)^{1,✉}, [A Ruya Yazici](#)¹

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PMCID: PMC11026205 PMID: [38635051](https://pubmed.ncbi.nlm.nih.gov/38635051/)

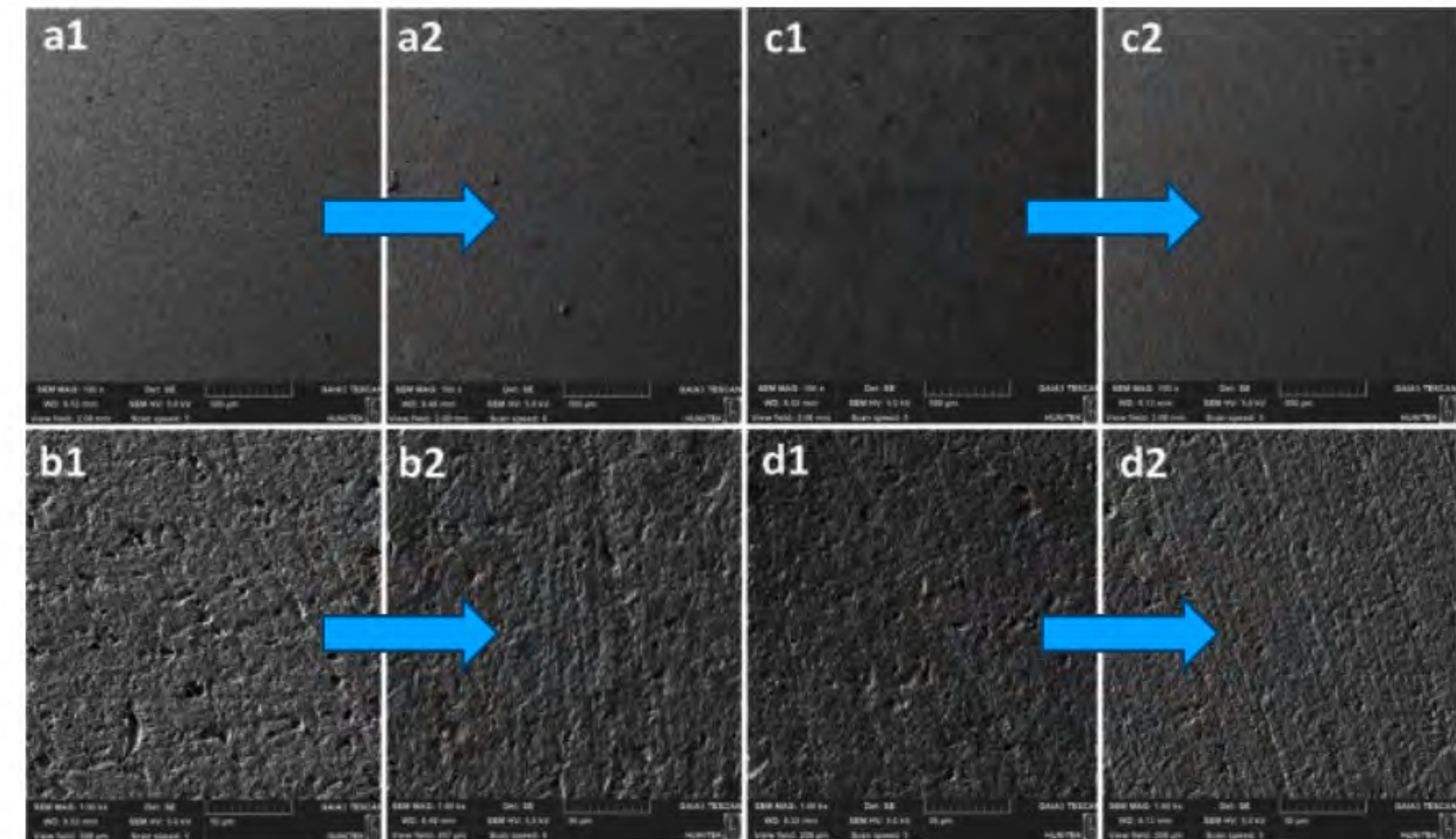


Fig 2. Bioactive Resin Composite





Case #1

Noah Shepard



- 58 y/o Latino Male
- Dx of **Oropharyngeal Squamous Cell Carcinoma**
- **35** total radiotherapy sessions totaling **70 Gy**, as well as a boost of **50 Gy** to the surrounding area
- **3** cycles of chemotherapy
- Patient consumed two cans of alcohol and 1 pack of cigarettes daily





Dental Exam 6 months post-therapy showed:

- Dry oral mucosa
- Hyposalivation
- Gingival Inflammation
- Biofilm Accumulation
- Soft, light colored lesions in the cervical region
- Hard, dark colored lesions in the incisal region
- Failed MOD restoration on tooth #25





Material of choice for the cervical radiation caries was **RMGI**

- Caries removed with round latch bur at slow speed
- Polyester strip placed to avoid proximal excess
- RMGI placed with Centrix delivery system



Failed MOD restoration removed on #25

- Impression created using condensation silicone
- Mold created with the same material to obtain a semi-rigid model
- Restoration was then prepared over the model with a **bioactive resin composite** (Beautifil A3)
- Restoration prepared with aluminum oxide sand blasting
- Preparation prepared with pumice
- Restoration cemented with a **self-adhesive resin cement** (BeautiCEM), then polished



RMGI

Semi-Direct Resin Composite

Microleakage & Sensitivity

-Superior control of microleakage compared to flowables; reduces post-op sensitivity

-Enhanced marginal adaptation minimizes microleakage
-Reduced through shorter procedures

Adhesion & Bonding

-Excellent chemical and micromechanical adhesion
-Increases bond strength over time

-Good marginal adaptation
-Relies on adhesive systems for bonding

Fluoride Release

-Sustained fluoride release promotes remineralization,
-Anti microbial effects

-Limited or none

Polymerization shrinkage & Stress

-Low elastic modulus attenuates shrinkage stress

-Reduced shrinkage stress due to extra-oral pre-polymerization

Durability & Load Bearing

-High flexural strength, less brittle
-Suitable as a liner
-Not ideal for high-load areas alone

-Superior resistance to occlusal loads in posterior restorations



Behavioral Objective 4

Identify the most commonly used restorative materials in recent years for irradiated patients.

Megan Black

Search Strategy



Databases used

- Pubmed

P: Patients with head and neck radiation therapy or irradiated teeth

I: Restorative materials, especially composite resins and adhesive systems

O: Material choice, clinical performance, success rate, bond strength, etc.

MeSH Terms

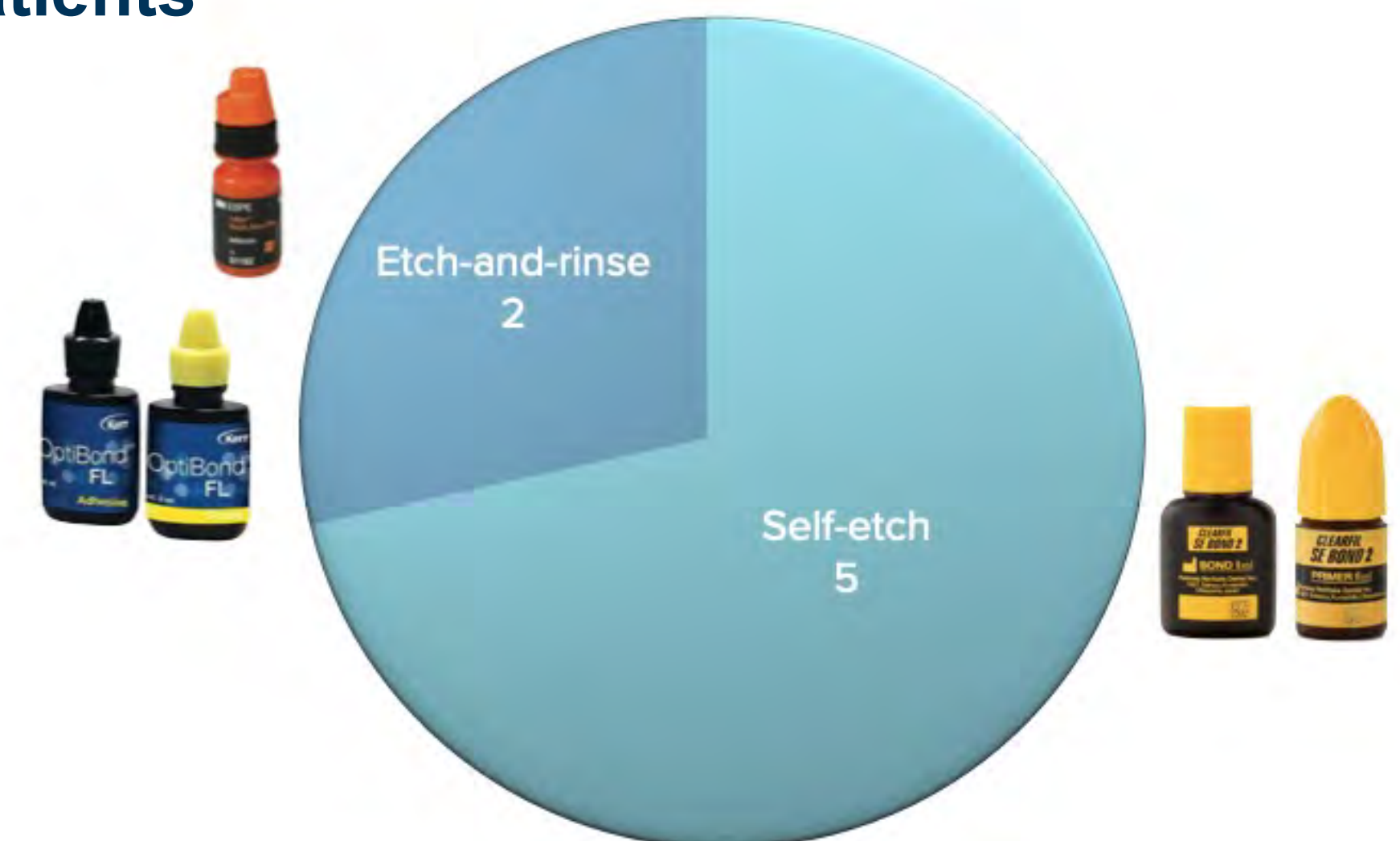
- ("irradiated teeth" OR "oral cancer radiotherapy") AND ("composite resin" OR "dental adhesive" OR "restorative material" OR "glass ionomer")

Filters

- Last 5 years, review

Highest level of evidence

- Systematic Review





Category	Materials Mentioned	Commercial Brands	Number of featured studies
Composite Resin	Resin-based composites	Filtek Z350 (3M ESPE), Herculite XRV (Kerr)	12
Glass Ionomer Cement (GIC)	Conventional glass ionomers	Ketac-Molar Aplicap (3M ESPE), Fuji IX GP (GC), Ketac-Fil (3M ESPE)	6
Resin-Modified GIC (RMGIC)	Hybrid resin–glass ionomer	Photac-Fil (3M ESPE)	1
Adhesive Systems	Etch-and-rinse and self-etch adhesives	Optibond FL (Kerr), Adper Scotchbond MP Plus (3M ESPE), Adper Single Bond 2 (3M ESPE)	4
Amalgam	Metal-based restorative material	—	1



Behavioral Objective 5

Apply a clinical decision tree to **select** appropriate restorative materials and preventive strategies for patients undergoing or having completed head and neck radiation therapy, tailored to their risk factors and clinical presentation.

Lizzie Winders

Comprehensive Pre-Radiation Dental Assessment

Advanced caries
Periodontal health
Radiographs



Patient Risk Factors

Type of radiation (IMRT vs conventional)
Oral Hygiene
Salivary flow (post-treatment)

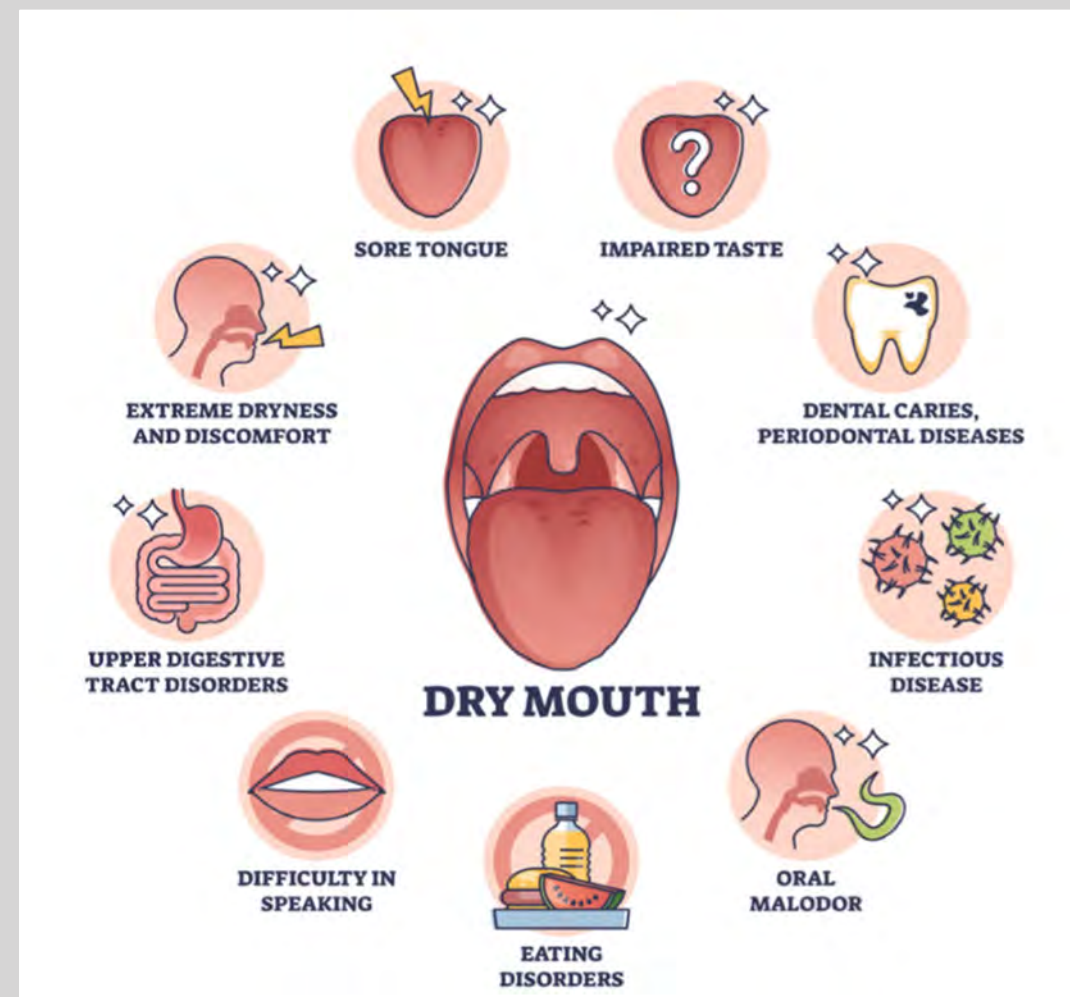


Then address any radiation induced issues of the oral mucosa...



Xerostomia

- Pilocarpine → prescription required
- Chlorhexidine for caries prevention
- Xylitol gum

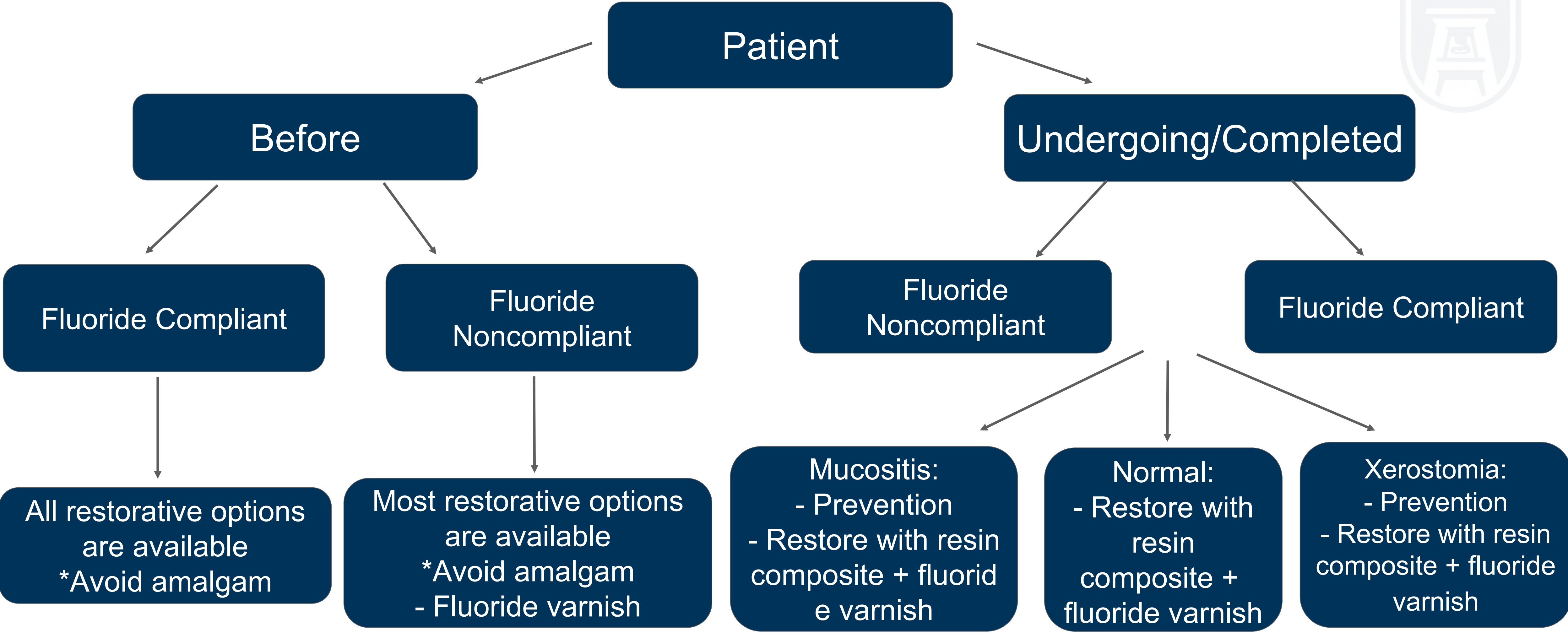


Mucositis

- Alcohol free products
- Corticosteroids
- Magic mouthwash



Now how to decide what restorations to use...



Best restorative material is resin composite in addition to routine fluoride varnish applications every 3 months

In Summary



- Choose material based on patient risk and tooth condition
- For post-radiation caries → use **resin composite + fluoride**
- GIC = backup option for low fluoride users, but fails faster
- **Avoid amalgam** and consider removing any old amalgam before radiation
- Avoid extractions/implants in irradiated bone if possible
- Keep restorations simple and **minimally invasive**



Other Considerations....



- Bioactive materials (materials that interact with soft tissue to promote regeneration and remineralization) have potential promise in benefiting these patients
 - Ongoing research
 - Evidence is not strong enough to confirm or deny its benefit to irradiated or nonirradiated tissue
- There is much conflicting literature concerning our topic
 - More clinical evidence is needed to support these claims
 - Randomized control trials, in vivo studies to mimic the oral environment, uniformity among the composition of materials tested...





Case #2

Tomi Adeogun

- 53 y/o white woman
- CC: “I had cancer treatment... now my teeth are brown.”
- Previously diagnosed with **Stage I OSCC** in 2020
- Treated with surgical excision and conventional HNRT of **50 Gy** over 6 weeks.

Dental exam reveals she has **radiation-related caries on #6-11.**



**Will you restore with
resin composite or
RMGI?**



Key Patient Factors:

- Salivary flow: Moderately reduced
- Fluoride compliance: Highly receptive to using fluoride trays, receiving in-office treatments
- Oral hygiene: Brushing 2x/day with OTC F- toothpaste, uses F- mouthrinse

Material Criteria and Indications

	Resin Composite	RMGI
Moisture tolerance	Sensitive	Tolerant
Durability with Xerostomia	More stable	Higher degradation
Fluoride release	None	Continuous
Pt F- management	For compliant pts	For noncompliant pts
Esthetics	Excellent	Moderate





Key Patient Factors:

- Salivary flow: Moderately reduced
- Fluoride compliance: Highly receptive to using fluoride trays, receiving in-office treatments
- Oral hygiene: Brushing 2x/day with OTC F- toothpaste, uses F- mouthrinse

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Esthetics	Excellent	Moderate





Decision?



Restore lesions with Resin composite

- Ensure proper adherence to fluoride treatment and oral hygiene



Behavioral Objectives

BO1. Identify the oral and systemic effects of head and neck radiation therapy, including its impact on teeth, bonding substrates, salivary function, and overall restorative prognosis.

Head and neck radiation therapy causes systemic effects such as **fatigue**, **weight loss**, and **immunosuppression**. **Oral effects** include mucositis, candidiasis, ulcerations, dysphagia, and taste changes, which impair nutrition and healing. **Salivary gland damage** leads to xerostomia and radiation **caries**, while fibrosis and lymphedema can cause **trismus** and **reduced tongue mobility**. **Radiation decreases enamel hardness** and mineral content, making it **brittle and fracture prone**. In **dentin**, radiation alters microstructure, **degrades collagen**, increases surface roughness, and **reduces mechanical integrity**. These changes **lower bond strength** of adhesives, so **restorations are ideally completed before radiotherapy** to optimize longevity.

Behavioral Objectives

BO2. Differentiate between restorative materials (amalgam, composites, GIC, RMGIC, ceramics, zirconia, bioactive materials) by explaining how radiation may influence their clinical performance and interaction with oral tissues.

Radiation affects both the **mechanical properties and clinical performance of restorative materials**. **Amalgam and metals are stable and resist degradation** but can **require extensive tooth reduction** and may locally **increase radiation dose** due to radiation scattering. **Composite resins maintain good strength**, yet their **bond strength is reduced** after radiotherapy because of **collagen breakdown in dentin**. **GICs and RMGICs release fluoride and help prevent caries**, but *both deteriorate in xerostomic conditions*, resulting in rough, *plaque-retentive margins*. **Ceramics and zirconia are mechanically strong and durable, but for irradiated teeth, there is no clinical evidence yet**. Bioactive materials may support remineralization, though *long-term clinical data in irradiated teeth remain limited*.

Behavioral Objectives

BO3. Compare and contrast the advantages and limitations of restorative material choices based on scientific evidence in the context of radiation-induced changes such as xerostomia, tooth brittleness, and altered bonding.

Conventional GIC releases fluoride, which helps prevent radiation-related caries, but is *brittle and mechanically less durable*. RMGIC improves strength and fluoride release, yet still *deteriorates under xerostomic conditions and remains less durable than composites*.

Composite resins offer superior long-term survival, mechanical properties, and resistance to hydrolytic and enzymatic degradation. *Bioactive composites* may enhance remineralization and tissue preservation, though *clinical evidence in irradiated patients is limited*. Regarding adhesives, *self-etch systems are gentler on irradiated dentin and provide both chemical and micromechanical bonding*, whereas *etch-and-rinse adhesives may achieve higher bond strength on non-irradiated dentin but can overly demineralize fragile irradiated collagen, increasing risk of bond failure*.

BO4. Identify the most currently commonly used restorative materials for irradiated teeth.

The ***most commonly used restorative materials were resin-based composites***, mainly Filtek Z350 (3M ESPE) and Herculite XRV (Kerr), conventional GICs like Ketac-Molar Aplicap (3M ESPE) and Fuji IX GP (GC), and **adhesives** such as Optibond FL (Kerr) and Adper Scotchbond MP Plus (3M ESPE), with **self-etch systems** showing ***better performance on irradiated teeth.***

BO5.Apply a clinical decision tree to select appropriate restorative materials and preventive strategies for patients undergoing or having completed head and neck radiation therapy, tailored to their risk factors and clinical presentation.

Before radiotherapy, most restorative materials and adhesive systems can be used, as they are **not significantly affected by radiation**. Exceptions are extensive preparations for ceramic or metal crowns, which are generally not recommended due to poor prognosis in irradiated patients. *During radiotherapy, elective dental work should be avoided, focusing instead on optimal oral hygiene. After radiotherapy, in patients with xerostomia, self-etch adhesives combined with resin composites are preferred, as they are gentler on irradiated dentin and provide both chemical and micromechanical adhesion.* Preventive maintenance should include daily 0.05% NaF rinse, saliva substitute, and recall visits every 3–6 months. GIC or RMGIC can be considered for high-risk teeth for their fluoride release, though mechanical limitations *must be recognized*.



BO 1



BO 2



BO 3



BO 4



BO 5



Thank You!

Restorations in Radiation: Material Choices for Challenging Cases

Presented by: Mildred Bonful • Doha Akid • Aidan Rackliffe • Daniel Lee
Megan Black • Lizzie Winders • Tomi Adeogun • Noah Shepard

Faculty Mentor: Dr. Gabriela Magalhaes

“Balancing science, skill, and sensitivity — one restoration at a time.”



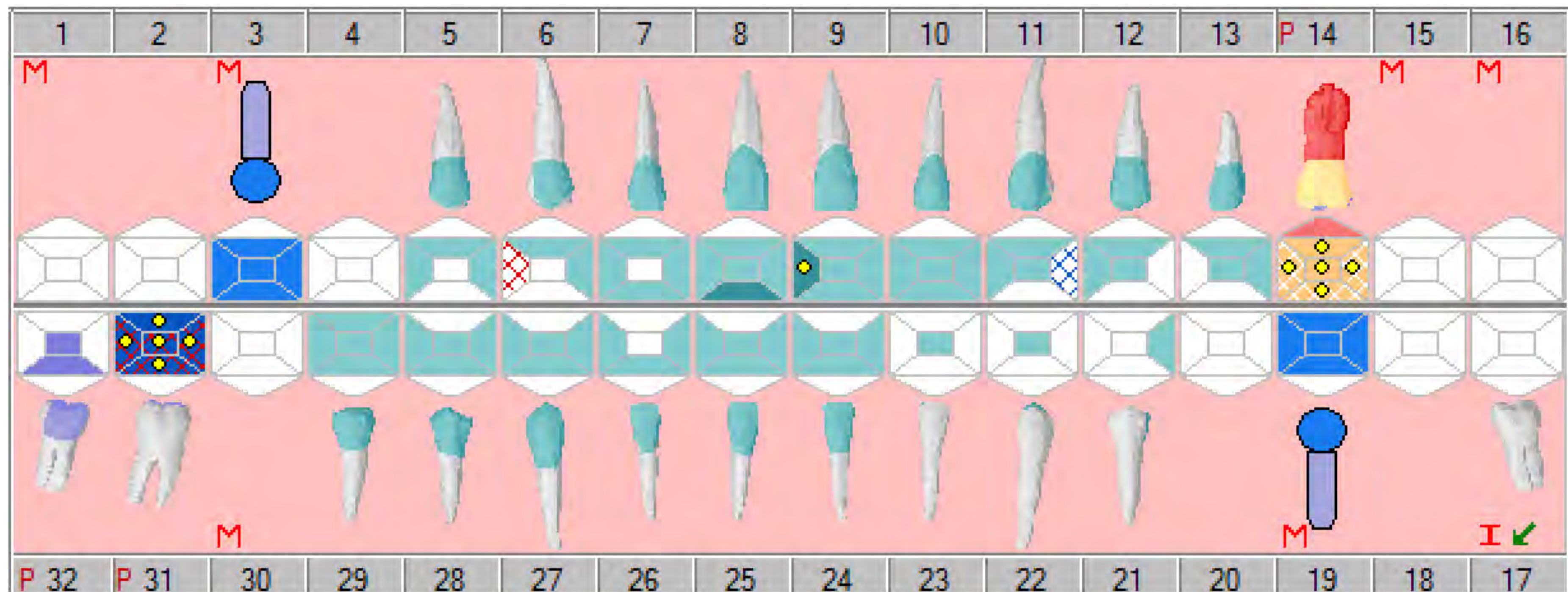
Six Thinking Hats®



Six Thinking Hats®



Red Hat (Patient): Expresses emotions, concerns, and preferences regarding the material.



"I have had to get so many fillings done since my treatment...but even worse is the number of times I have had to get the **same fillings redone**." - Patient

Six Thinking Hats®



Black Hat (Dentist): Evaluates potential risks, challenges, and clinical implications.

Risks and potential challenges:

- How long will this restoration last before secondary decay?
- How will I manage my patient's expectations for the longevity of certain materials?
- How will I emphasize the importance of fluoride for the long term success of restorative materials and prevention of decay in general?

Clinical Implications:

- Resin composite with fluoride varnish is our best option for any patient undergoing radiotherapy or post-radiotherapy
- HOWEVER...it is still not a perfect solution
- These patients require continual follow-up and close monitoring due to lifelong high caries risk



Six Thinking Hats®



White Hat (Faculty): Provides objective data and scientific evidence about the dental material.

“Head and neck radiation makes enamel more brittle and demineralized, denatures dentin collagen, and weakens the hybrid layer. That means our traditional bonding is less predictable — bond strength drops, and the risk of restoration failure goes up.”

“In irradiated patients, composite resin restorations show lower failure rates and better survival than traditional glass ionomer or RMGI. Glass ionomer and RMGI do release fluoride, but they break down faster in a chronically dry, xerostomic mouth. ‘Bioactive’ materials are interesting, but they still don’t have long-term clinical proof. Right now, composite resin should be considered the primary restorative material of choice for irradiated teeth.”

Six Thinking Hats®



Yellow Hat (Sales Rep): Highlights the benefits and advantages of the material.

"Our bioactive resin continuously releases calcium, phosphate, and fluoride ions to **strengthen the surrounding dentin** — ideal for patients with radiation-related demineralization."

"Even in dry oral environments, these bioactive adhesives help buffer pH and support remineralization — giving your restoration a **therapeutic edge.**"

Six Thinking Hats®



Green Hat (Manufacturer): Explores innovation, and future possibilities for the material.

"Post-radiation patients may only make up a small proportion of the entire population that we serve, but we value all patients the same. We strive to go above and beyond to understand the unique challenges faced by patients who've triumphed over radiation therapy. Those who went from once vibrant smiles, that now contend with heightened sensitivity and enamel vulnerability. Our materials aren't just fillers, they're lifelines. Together we are restoring more than teeth; we're restoring joy with every new innovation."

Six Thinking Hats®



Blue Hat (Board of Dentistry): Ensures all perspectives and standards are considered.

“The standard of care here is evidence-driven, not marketing-driven. For an irradiated patient with compromised bonding and extreme caries risk, the dentist is responsible for selecting and justifying a material that will last, using preventive strategies, and minimizing tooth reduction. This goes beyond placing a restoration — it’s protecting the patient’s long-term oral function and quality of life.”